

A retrieved-context account of selective interference in intrusive and voluntary remembering

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Abstract

In the trauma-film paradigm, a visuospatial task after a later film reminder can reduce the frequency of film-related intrusions while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that involuntary and deliberate remembering depend on different memory representations. We propose instead that the dissociation can reflect differences in how retrieval is cued and controlled. We implement this account in an emotional Context Maintenance and Retrieval (eCMR) model that simulates film, reminder, and task encoding and predicts sequences of film and non-film recall events. In the model, a film reminder reinstates film-associated context before the task. Task items are then encoded in association with that context, so later film-related context can cue task competitors as well as film items. Deliberate recall is less affected when a maintained film-retrieval goal gives film candidates an additional source of support without equally supporting task competitors. In tasks like recognition, the test presents the item, so the model evaluates the item's fit to context instead of selecting among film and task candidates. This comparison can therefore make voluntary memory appear spared because of test format rather than because voluntary remembering is generally unaffected. Interpreting selective interference this way motivates better-matched tests of intrusive and voluntary remembering and situates trauma-film findings within a broader computational account of episodic retrieval.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

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Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. From a functional perspective, events with lasting significance need to remain accessible in memory so remembered details can guide later decisions (Zhou et al., 2025). At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current goals, cues, and circumstances. Intrusive memories after trauma suggest that durable memory can strain the selectivity of retrieval. In PTSD, such memories can come to mind unbidden and poorly matched to the present, causing distress and interfering with ongoing activity (Brewin et al., 2010; Ehlers & Clark, 2000; Iyadurai et al., 2019; Krans et al., 2009). Explaining how episodic memory preserves consequential events while regulating their retrieval is therefore a shared target for both clinical science and mechanistic theories of memory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event (Holmes & Bourne, 2008; James et al., 2016). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to come to mind involuntarily. In such research, intrusion frequency can be weakly related to voluntary-memory performance, such that the two measures dissociate (Holmes et al., 2004; James et al., 2016). The same encoded episode can therefore support both involuntary and deliberate access, and these modes can come apart empirically.

Visuospatial interference tasks make this dissociation experimentally sharper. In immediate post-film studies, tasks such as Tetris can reduce later film intrusions while voluntary memory for the same material remains relatively intact (Deeprrose et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). Delayed variants add a later reminder before the interference task. Participants return after roughly 24 hours to 3 days, view reminder cues for the film, and then complete Tetris. These reminder-plus-Tetris procedures can also reduce subsequent

intrusions (James et al., 2015; Kessler et al., 2020). Related procedures have been extended beyond analogue trauma films to real-life and occupational trauma, including motor vehicle accidents, emergency caesarean section, refugee trauma, and health-care work during COVID-19 (Beckenstrom et al., 2026; Horsch et al., 2017; Iyadurai et al., 2018, 2023; Kanstrup et al., 2021, 2024; Ramineni et al., 2023). At the same time, recent reviews and replication attempts caution against treating the effect as settled or uniform (Asselbergs et al., 2023; McConnell et al., 2026; Varma et al., 2024; Wessel et al., 2025). This selective interference effect therefore raises a theoretical question: when and why can an intervention alter intrusion-like and voluntary access differently, even when both concern the same recent experience? Figure 1 schematizes this target as an interaction among reminder timing, task condition, and retrieval condition.

The selective interference effect has been used to argue that intrusive and voluntary remembering depend on differently vulnerable memory representations (Brewin et al., 1996; Brewin, 2014). Dual-representation accounts give this interpretation an architectural basis. Intrusive re-experiencing is linked especially to sensory-bound representations, whereas deliberate remembering depends more on contextualized episodic representations (Brewin et al., 2010; Brewin, 2014). In immediate post-film interventions, visuospatial tasks are understood to compete for working-memory resources while sensory-perceptual imagery from the film is still being maintained and consolidated (Baddeley & Andrade, 2000). Because later intrusions are assumed to arise especially from sensory-bound representations, this competition should reduce intrusions while leaving voluntary-memory performance relatively preserved (Deeprase et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). In delayed post-reminder interventions, the reminder is instead treated as reactivating the relevant memory before the task, allowing the interference procedure to alter later availability through reconsolidation or post-reactivation updating (James et al., 2015; Kindt et al., 2009). Across these variants, selectivity is explained as a stronger effect on the memory basis for intrusions than for voluntary remembering.

Here we argue that preserved voluntary memory does not, by itself, establish that a separate representation was spared. Selective interference can instead arise from differences in

how film memories are cued and controlled at retrieval. In the account developed below, a film reminder reinstates context associated with the film before task material is encoded. Task material then becomes linked to that context. When later remembering is guided by film-related context, task and film material can therefore compete, interfering with film recall. The task interferes less when retrieval depends less on context to access film material. Deliberate recall can be less affected when a maintained film-retrieval goal gives film candidates an additional source of support without equally supporting task competitors. Tasks like recognition can also be less affected, because the test presents a candidate film item for judgment rather than requiring film material to be brought to mind. On this account, the selective-interference effect emerges when retrieval conditions differ in how much they depend on film-related context that also cues task material.

We formalize this retrieval account in an emotional variant ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)) of the Context Maintenance and Retrieval (CMR) model ([Polyn et al., 2009a](#)). Retrieved-context models treat context as a gradually changing state that is shaped by experience, reinstated by retrieval, and used to cue later memory search ([Howard & Kahana, 2002](#); [Polyn et al., 2009a](#); [Sederberg et al., 2008](#)). In CMR, items and context are linked by bidirectional associations. Context cues candidate items, and retrieved items or items presented at test reinstate associated context ([Polyn et al., 2009a, 2009b](#)). Emotional CMR variants retain this item-context architecture while allowing emotional or source context to shape what is bound and later retrieved ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)). These properties make CMR a principled setting for a retrieved-context account of selective interference based on context reinstatement and item-context binding.

Applied to the trauma-film paradigm, the model treats film, delay-period, task, and filler events as a continuous sequence through which context evolves. A film reminder is implemented as item-to-context reinstatement before task encoding. Later recall is implemented as context-to-item sampling from the film and non-film items supported by the current retrieval state. When that context supports both film and task material, task items compete with film recall rather

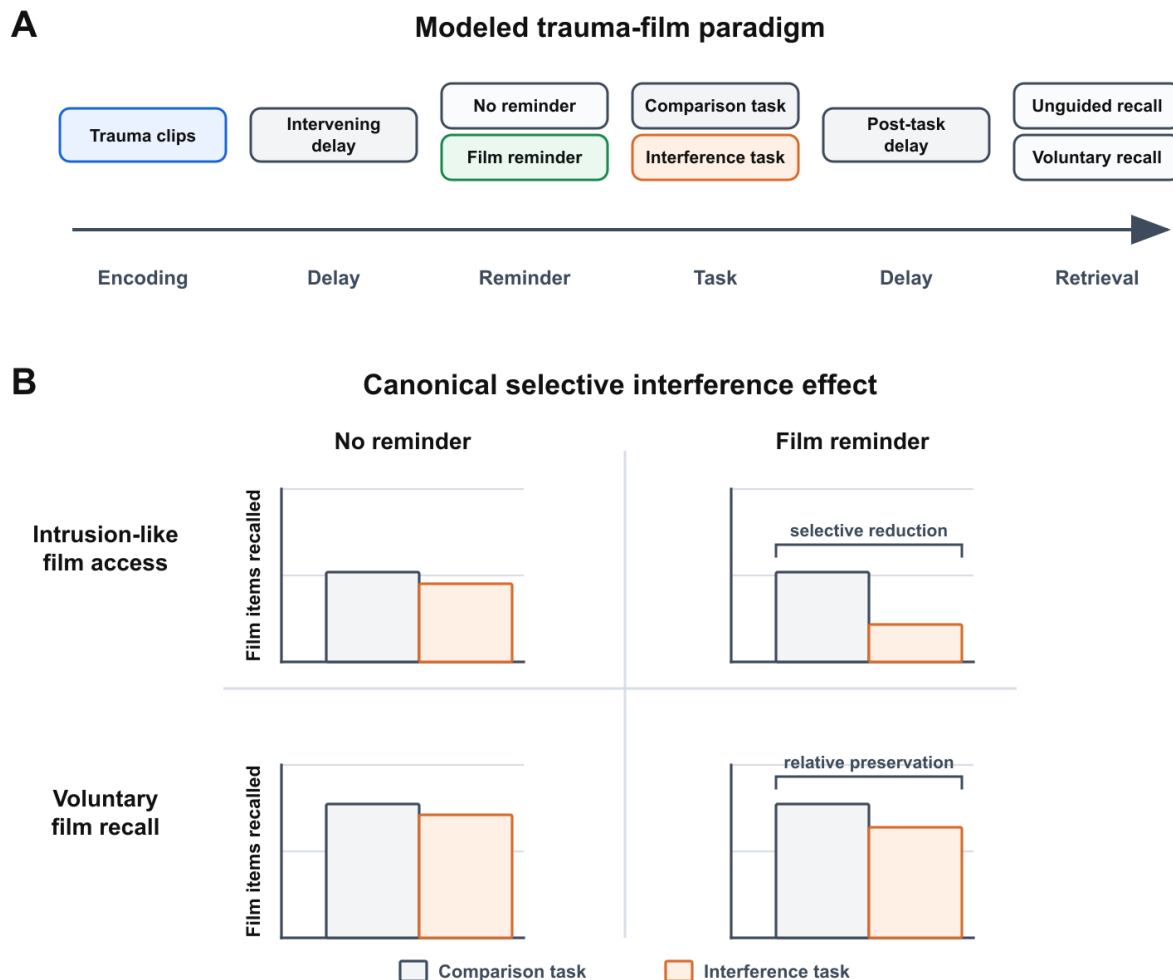
than merely weakening a film trace.

Deliberate recall is modeled with several retrieval-control processes. Rather than universally protecting film recall in a manner consistent with the selective interference effect, we show that some control mechanisms can increase film recall without making recall less sensitive to reminder-linked task encoding. In the simulations, the strongest protection comes from adding a maintained film-retrieval goal that gives film candidates support not shared by task competitors. This identifies a functional condition for selective interference: voluntary recall is protected when it depends less on film-associated temporal context that has also become supportive of task material.

The simulations follow a generalization-style modeling logic ([Busemeyer & Wang, 2000](#); [Lohnas et al., 2023](#)). We anchor core eCMR parameters in an independent affective free-recall dataset ([Dupertuys et al., 2026](#)), then map the trauma-film sequence onto that architecture and ask whether it produces the qualitative selective-interference pattern. The first simulations examine whether reminder-linked task encoding creates competitors for film recall. The next simulations ask when deliberate recall is protected from those competitors, focusing on whether a maintained film-retrieval goal gives film material a more selective basis for recall. A final recognition analysis treats recognition as a boundary condition: when a candidate film item is supplied for judgment, performance need not depend on sampling that item from a context that also cues task material. Together, the simulations define a mechanistic alternative to selective-vulnerability accounts and identify conditions under which intrusive and voluntary remembering should appear more or less dissociated.

Figure 1

Paradigm variables and canonical selective interference effect. (A) The modeled trauma-film paradigm includes film encoding, an intervening delay, a possible pre-task film reminder, an intervening task, a post-task delay, and a later test of unguided or voluntary recall. (B) The canonical selective interference effect is summarized as a schematic interaction among reminder condition, intervening task, and retrieval condition. Columns distinguish whether the task is preceded by a film reminder, rows distinguish intrusion-like film access from voluntary film recall, and paired bars compare film output after a comparison task versus an interference task. The interference task produces the largest reduction in intrusion-like film access when it follows a film reminder, whereas voluntary film recall shows a smaller reduction under the same conditions. Bars are schematic and do not plot empirical counts, fitted values, or model output.



Retrieval Conditions in Selective Interference

Figure 1 summarizes the selective interference effect as reduced intrusion-like access with relatively preserved voluntary memory for the same film material. Studies of this effect use tests that differ in whether film content must be generated or is presented for judgment. Diary measures attempt to count film memories that inadvertently arise during daily life. Laboratory intrusion tasks count memories reported during a defined monitoring period, sometimes while film cues are present. Recognition tests present candidate film statements or still images for judgment. Free recall and retrieval probe tasks require participants to produce or elaborate film details from a prompt (Holmes & Bourne, 2008; James et al., 2016; Lau-Zhu et al., 2019, 2021). The resulting comparisons differ in whether film content is already supplied or has to come to mind.

Many studies that established selective interference used recognition as the voluntary-memory test. Immediate post-film Tetris studies and delayed reminder-plus-Tetris studies report fewer diary or laboratory intrusions while verbal or visual recognition remains similar across groups (Holmes et al., 2009, 2010; James et al., 2015; Kessler et al., 2020). Later diary-recognition work uses the same basic pairing (Asselbergs et al., 2024). Those results show that the intervention can reduce film memories reported as intrusions while leaving participants able to recognize candidate film material at test. Later studies made the concern more specific: diary reports depend on film memories coming to mind, whereas recognition tests supply candidate film material for judgment (Lau-Zhu et al., 2019, 2021). Preserved recognition is therefore narrower than general preservation of voluntary memory. Reviews, replication attempts, and newer tests also caution against treating the effect as uniform (Asselbergs et al., 2023; Hemi et al., 2023; McConnell et al., 2026; Varma et al., 2024; Wessel et al., 2025).

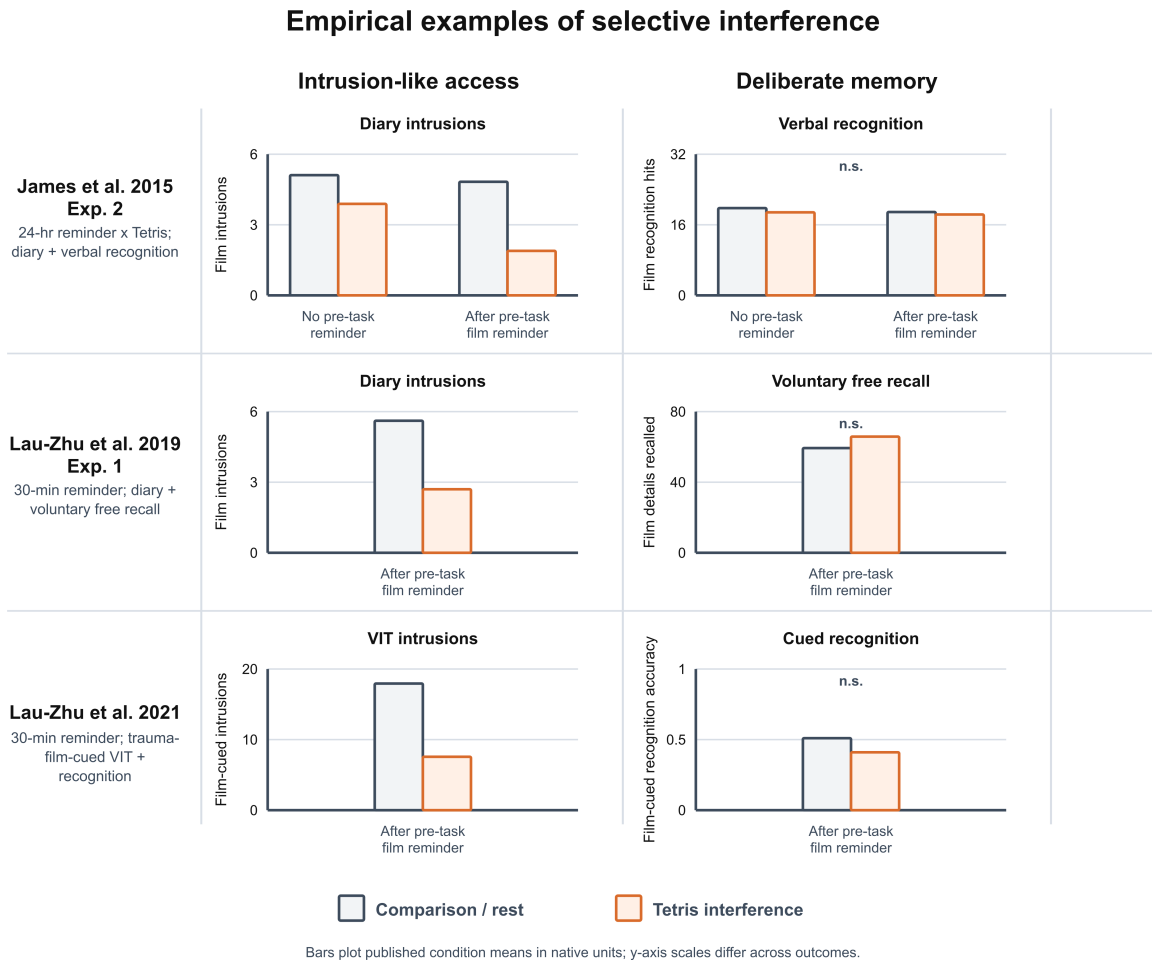
Studies using recall, probing, or vigilance-intrusion measures test whether the dissociation persists when film material has to be produced or reported under more closely matched conditions. One three-experiment study added free recall, specific probing, and priming to the diary-recognition comparison, then introduced a vigilance-intrusion task to measure intrusions during a laboratory monitoring task (Lau-Zhu et al., 2019). Reminder-plus-Tetris reduced diary

and vigilance intrusions while free recall, specific probing, and recognition were relatively preserved. Later work matched vigilance-intrusion and modified recognition measures more closely in cueing, context, timing, and attentional demands, again finding reduced intrusion measures with preserved recognition ([Lau-Zhu et al., 2021](#)). A recent preregistered study pushed the matching further by using voluntary and involuntary versions of the same vigilance retrieval task, but did not find the predicted reduction in involuntary memory ([McConnell et al., 2026](#)). Vigilance-intrusion tasks bring intrusion measurement into the laboratory. Participants perform an ongoing task and report film memories that arise, sometimes while film-related cues are presented. These cues can prompt film-related retrieval, but they do not supply the candidate film item for recognition. Vigilance-retrieval variants extend this logic by using similar task structure for voluntary and involuntary reports, changing the retrieval instruction while keeping cue timing, task context, and response format more closely matched ([McConnell et al., 2026](#)).

Figure 2 illustrates this empirical range with three exemplar designs: delayed-reminder diary-plus-recognition, delayed-reminder diary-plus-free-recall, and trauma-film-cued vigilance-intrusion-plus-recognition.

Figure 2

Empirical examples of selective interference across retrieval formats. Rows show three exemplar studies that differ in how intrusion-like and voluntary memory are measured: a delayed-reminder diary-plus-recognition design, a delayed-reminder diary-plus-free-recall design, and a trauma-film-cued vigilance-intrusion-plus-recognition design. Left panels plot intrusion-like film access and right panels plot deliberate memory. Gray bars indicate comparison or rest conditions and orange bars indicate Tetris interference. Bar heights reproduce published condition means in native study units, so y-axis scales differ across outcomes.



Panel A of Figure 1 defines the target paradigm for the trauma-film recall simulations: film encoding, a possible reminder, task encoding, and later unguided or deliberate retrieval that can

produce sequences of film and non-film recall events. The unguided retrieval condition approximates laboratory intrusion measurement rather than recognition. Film-related cues can be present during a monitoring task, but the film item still has to be sampled and reported if it comes to mind. Recognition is a boundary condition because the candidate film item is already present at test. When a test supplies the item for judgment, memory can succeed through an item-to-context route rather than through sampling from the current retrieval state. The retrieved-context account therefore has to explain why reminder-linked task material can compete with film recall during vigilance-intrusion retrieval while recognition remains relatively preserved.

A Retrieved-Context Account of Selective Interference

Figure 3

Encoding and associative structure in the retrieved-context account. (A) The model stores bidirectional item-context associations. Context-to-feature associations cue candidate items from the current context, and feature-to-context associations reinstate context when an item is recalled or when a reminder or test cue is presented. (B) The interference manipulation changes which context states task items are linked to during encoding. Without a pre-task reminder, task items are encoded after context has drifted away from the film. With a pre-task film reminder, task encoding occurs after film-associated context has been reinstated, so task items are learned in context states that overlap with film context. Thus, the same context that can cue film items can also cue task competitors during later retrieval. When task encoding is strong, those task items become stronger competitors during later context-guided retrieval. Circles mark item features: *F* labels film items and *T* labels task items. Squares mark context states. Blue marks film items or film-linked context, orange marks task items or task-linked context, green marks reminder cues, and gray marks neutral, delay, comparison, or shared model structure.

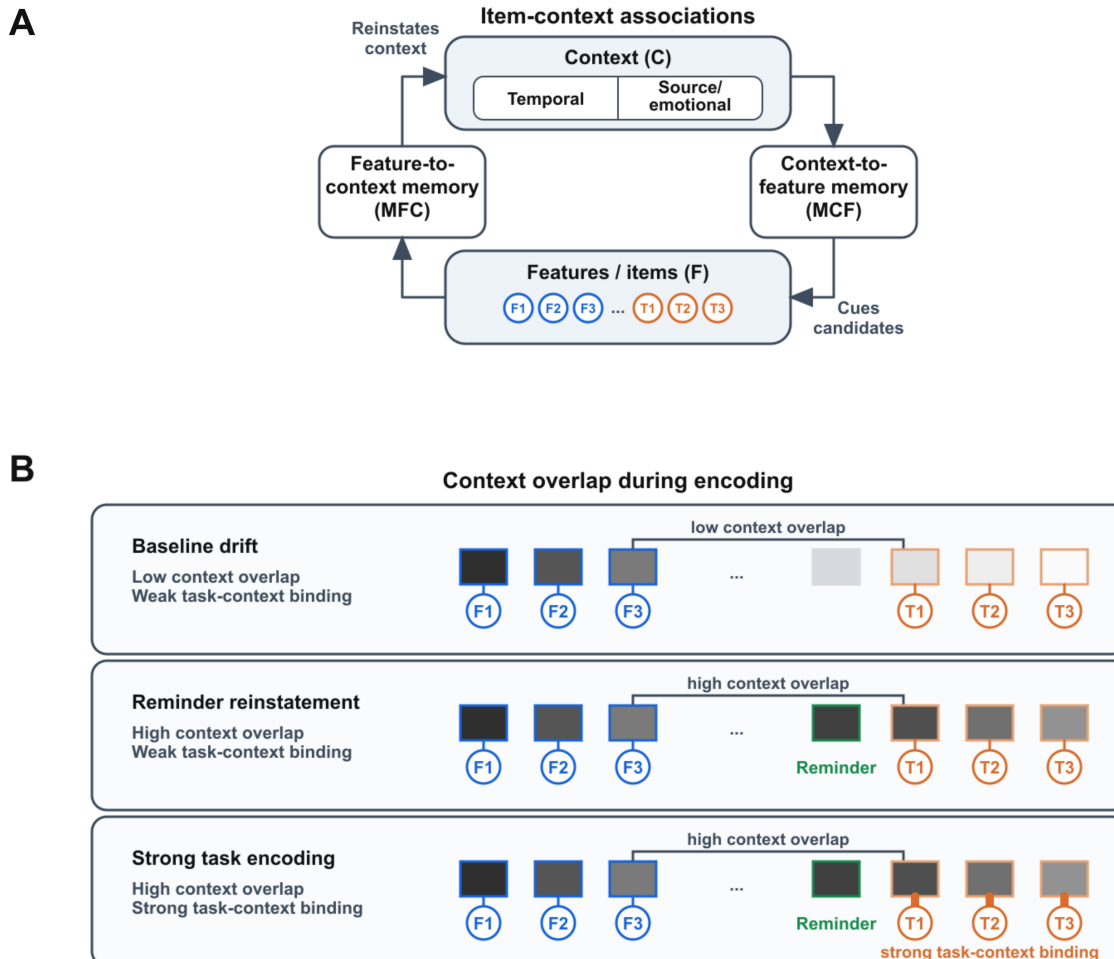
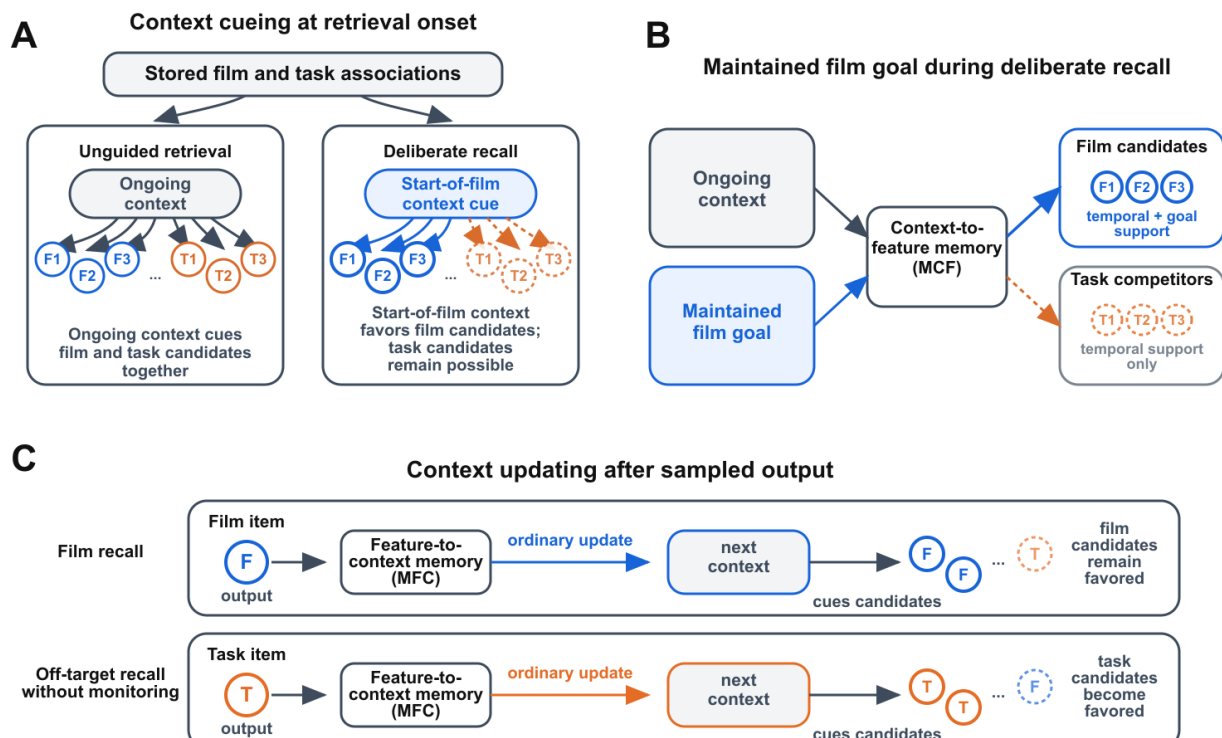


Figure 4

Retrieval operations in the retrieved-context account. (A) The upper box represents stored film and task associations; the lower boxes compare the context state used to begin unguided retrieval and deliberate recall. Stored film and task associations can be sampled from different starting contexts. Unguided retrieval begins from ongoing context, which can support film and task candidates together. Deliberate recall can begin with start-of-film context reinstatement, which favors film candidates while leaving task candidates possible. (B) A maintained film goal provides an additional deliberate-retrieval signal during context-to-feature sampling. Ongoing context can still cue both film items and task competitors, but the maintained goal adds support to film candidates rather than to task competitors. (C) Rows compare film recall, unmonitored task recall, and monitored task recall. Sampled outputs update the next retrieval context through feature-to-context memory. Recalling a film item keeps later search biased toward film candidates. Recalling a task item without monitoring can shift later context toward task candidates. With retrieval monitoring, the task-item update is dampened, preserving some film-candidate support. Blue marks film items, film-biased context, or stronger film support; orange marks task items, task-biased context, or stronger task support; gray marks shared context and associative operations. Dashed outlines or arrows mark weaker support or dampened updating, not unavailable candidates.



Retrieved-context models explain recall as search from a changing context state shaped by recent experience and prior retrieval. The current context cues candidate items, and recalled items reinstate context from encoding, producing temporal organization in free recall (Howard & Kahana, 2002; Sederberg et al., 2008). Applied to selective interference, these operations explain how task material becomes linked to film-associated context and how later retrieval determines the extent to which that material competes with film material for recall.

Film, task, and delay periods are represented as sequences of events that update context. For the film, these events correspond roughly to scenes or segments. For the task and delay periods, they represent non-film events that can also be encoded and later sampled. A film reminder reinstates film-associated context before task encoding. Task events encoded in that state can later be cued alongside film material. Retrieval then depends on how strongly the test setting constrains search toward film material rather than toward the broader set of candidates supported by the current context. Figure 3 summarizes the encoding and associative structure of the account, and Figure 4 summarizes the retrieval operations.

Context-Guided Search and Source Organization

Retrieved-context models were developed to explain structured regularities in open-ended recall. Free recall favors recently studied items, and successive recalls tend to come from nearby study positions (Howard & Kahana, 2002; Kahana, 1996; Sederberg et al., 2008). In these models, a gradually changing context state cues candidate items, and recalled items reinstate the contexts in which they were encoded. Recency follows because the retrieval context remains similar to contexts from recent study and recall events. Temporal contiguity follows because a recalled item reinstates context from its study position, which is similar to the contexts associated with nearby items.

Recall is also organized by shared content, task, and source. Semantically related items, items encoded under the same task, and items from the same source can cluster during recall (Lohnas et al., 2023; Morton & Polyn, 2016; Polyn et al., 2009a, 2009b, 2012). CMR captures this broader organization by allowing semantic, task, and source information to contribute to the

context state that guides retrieval. Items that share these forms of context can support one another during memory search even when temporal position alone would not explain the organization. In an account of the trauma-film paradigm, this lets film material receive shared film-directed support while individual film scenes or segments remain distinct encoded events.

Emotional memory gives source organization a further role in the present model. Emotional items can cluster together, and emotional material is often remembered better than neutral material (Talmi et al., 2007; Talmi, 2013; Talmi & Moscovitch, 2004). Emotional CMR represents affective information as a source-like context dimension and allows emotional items to form stronger source-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). The same architecture can therefore give trauma-film material a shared film-directed signal and stronger retrieval support, while leaving neutral task and delay material available as separate competitors.

Reminder-Linked Competitor Encoding

Reminder-linked task interventions produce interference by changing what task events become associated with during encoding. In a retrieved-context account, task events interfere with film recall when they are encoded in context states that later also support film material.

A film reminder can change the context in which the later task is encoded. In retrieved-context models, presented or retrieved items can reinstate context from encoding (Howard & Kahana, 2002; Polyn et al., 2009a; Sederberg et al., 2008). When a film cue is presented before the task, it shifts current context toward states associated with the earlier film episode. Task events are then encoded in a context that can later also support film material.

Recall can include material outside the intended target. In multi-list free recall, participants sometimes report items from earlier lists instead of the list they were asked to recall (Howard et al., 2008; Lohnas et al., 2015; Zaromb et al., 2006). Those errors are structured: they are biased toward recently studied prior lists and change as intervening study or delay changes. Retrieved-context models explain this pattern by treating recall as candidate sampling from the current context. As context drifts, support for non-target items declines. When a non-target item was encoded in a recent or related context, it can still enter the candidate set. This makes task and

delay events eligible recall candidates in the trauma-film paradigm. If those events are encoded in context states that later support film recall, they can compete with film material during retrieval.

Delayed reminder-linked task interventions in the trauma-film paradigm instantiate this kind of competitor encoding. In these studies, participants encounter film cues before the interference task, and the procedure is typically interpreted as reactivating film memory before the task begins (James et al., 2015; Kessler et al., 2020). When participants return to an earlier learning context, new material can become linked to the earlier episode (Hupbach et al., 2008; Sederberg et al., 2011). The simulations instantiate this sequence by placing task encoding after reminder-driven reinstatement of film-associated context. Task events encoded in that reinstated state acquire associations with context that also supports film items. The intervention can then reduce later film recall by adding task competitors while stored film-item associations remain available. Panel B Figure 3 summarizes this encoding claim: the reminder shifts task encoding into film-overlapping context, so later context-guided retrieval can cue task competitors alongside film material.

Recognition and Supplied Film Items

The retrieved-context framework distinguishes two routes by which film material can enter a test response. In context-guided retrieval, the current state of context cues candidate items, so a film memory is reported only if film material is selected from the items supported by that state. In item-guided retrieval, a presented film statement or still image is already the candidate, so the task is to evaluate the context retrieved by that item. Trauma-film studies commonly assess voluntary memory with recognition tests, where participants judge whether presented film statements or still images appeared in the film (Lau-Zhu et al., 2019, 2021; Varma et al., 2024). By contrast, diary and laboratory monitoring tasks ask participants to report film memories that arise during daily life or during an ongoing task (Holmes & Bourne, 2008; James et al., 2016; Lau-Zhu et al., 2019). A retrieved-context account therefore interprets the dissociation between recognition and intrusion-like measures under reminder-linked task interference as primarily test-format dissociations. Comparisons between intrusion-like reports and recognition tests confound

retrieval intentionality with the route by which film material is accessed.

Recognition judgments are shaped by the context surrounding a test probe. Recently studied items are recognized more accurately than less recent items. Similar lures are harder to reject, and one recognition probe can influence recognition of items studied nearby (Healey & Kahana, 2016; Jin et al., 2024; Kahana, 2020; Schwartz et al., 2005). In retrieved-context accounts of recognition, a presented probe retrieves context associated with earlier encounters of that item, and the retrieved context is compared with the context active during the test (Healey & Kahana, 2016; Jin et al., 2024). Recency, lure similarity, and carryover between probes can be explained by changes in the match between the context retrieved by the probe and the context used for judgment. Recognition therefore probes memory with an item and evaluates the context that item retrieves.

Reminder-linked task encoding interferes with diary and monitoring measures when film-related context cues non-film competitors. Those measures count film material only when it is retrieved during daily life or during a monitoring period. After a film reminder, task events can be encoded while film-associated context is active. Later, film-related context can cue those task events along with film items, reducing the chance that film material is retrieved. Recognition uses the item-guided route described above. The tested film item can retrieve associated context without first being selected from film-related context. Figure 13 formalizes this supplied-item route: reminder-linked task items remain competitors during context-guided recall, but they do not compete for the initial availability of a presented film probe.

Voluntary recall can be tested by means other than recognition. In vigilance-intrusion tasks, participants perform an ongoing task while film or foil cues appear, and they report film memories that arise during the task (Lau-Zhu et al., 2019, 2021). A voluntary version can keep the same film-cued task structure while instructing participants to search deliberately for film memories, while an involuntary version can instead more neutrally instruct participants to report any film memories that come to mind (McConnell et al., 2026). Both versions score film material retrieved during a recall period rather than recognition judgments about supplied candidates.

Film cues in these tasks still shape retrieval. A film cue can reinstate context associated with the cued film material. If recall depends on maintaining film-related context across the retrieval period, film cues can support recall in both monitoring and deliberate-search conditions. That cue support does not automatically remove reminder-linked competition. After reminder-linked task encoding, the reinstated context may also support task material, enabling task items to compete with film material during context-guided recall. A recognition probe changes the access route more sharply because the probe is the candidate item to be judged.

Recognition-intrusion comparisons therefore make confounds test format and retrieval intentionality in measures of the selective interference effect. Better-matched recall designs keep film material dependent on context-guided retrieval across intentionality conditions, with film cues modeled as inputs to the retrieval context. Under that design, differences in the effects of reminder-linked task encoding on film recall can be attributed to the way deliberate recall controls the course of memory search.

Film-Directed Recall Control Under Matched Retrieval Conditions

Competitor encoding makes film recall vulnerable when retrieval is guided by context that supports both film and task candidates. Diary and vigilance-intrusion measures count film memories when they come to mind, often during ongoing activity or monitoring rather than after a specific film item has been supplied (Holmes & Bourne, 2008; Lau-Zhu et al., 2019, 2021). The most relevant voluntary comparison similarly requires film material to be called to mind during a recall period rather than judged as a supplied recognition candidate (Lau-Zhu et al., 2019; McConnell et al., 2026). Under this matched recall design, a film reminder can make task material accessible from the same retrieval state as film material by associating task events with film-overlapping context. Control processes can improve deliberate recall by orienting retrieval toward the film. They protect film recall from reminder-linked interference to the extent that they make retrieval less dependent on film-related context that also supports task material.

Within this matched recall setting, Figure 4 separates three ways deliberate recall can differ from unguided retrieval after the same reminder-linked task encoding. Start-of-film context

reinstatement acts before sampling by changing the initial context state. A maintained film-retrieval goal acts during sampling by adding support to film candidates. Retrieval monitoring acts after sampling by reducing the carryover from off-target candidates. These control points make separable predictions because they change different parts of the retrieval cycle. The simulations test which control changes the effect of reminder-linked task encoding, rather than simply increasing film recall under all conditions.

Deliberate recall can begin from a context state biased by retrieval instructions. When people are asked to recall an earlier studied list after studying a more recent one, their early responses are often already biased toward the instructed list rather than showing a gradual search backward through recent-list items (Healey & Wahlheim, 2024; Lohnas et al., 2015). Recall also shows a reliable tendency to start near the beginning of a studied list, especially when the usual end-of-list advantage is weakened by a delay (Healey & Kahana, 2014; Herrema & Kahana, 2024; Howard & Kahana, 1999). Start-of-list context reinstatement mechanisms account for these patterns by evolving the current context toward the state associated with the beginning of the target list before retrieval begins (Healey & Wahlheim, 2024; Kragel et al., 2015; Lohnas et al., 2015; Morton & Polyn, 2016). In our retrieval control simulations, start-of-film reinstatement drifts the retrieval state toward the context present at the beginning of film encoding. This starting state favors early film candidates before ordinary context-guided sampling resumes, so its predicted benefit is tied to the temporal region that has been reinstated.

Retrieval monitoring acts after a candidate has been sampled. Evidence for this post-sampling control comes from externalized recall tasks where participants report every word that comes to mind and mark errors. These instructions reveal many more intrusions than standard recall, and many sampled intrusions are marked as errors, indicating that off-target candidates can enter retrieval without being accepted as target memories (Lohnas et al., 2015; Unsworth et al., 2010). Monitoring also carries forward across retrieval: when people fail to identify one response correctly, they are worse at rejecting the next intrusion (Unsworth et al., 2010). At the same time, sampled intrusions can pull search away from the target list, as intrusions tend to cluster with

other intrusions from related prior-list or semantic contexts ([Unsworth et al., 2010](#); [Zaromb et al., 2006](#)). Retrieved-context implementations have treated monitoring as a target-match filter while still allowing rejected items to update the next retrieval context ([Lohnas et al., 2015](#)). Our trauma-film simulations explore whether that post-sampling evaluation can also reduce carryover from sampled non-film candidates. A sampled non-film candidate still counts as a retrieval attempt, but its context update is dampened when it is evaluated as off-target. Monitoring can therefore help deliberate recall return to film material after non-film competition appears, while leaving the initial competition from reminder-linked task encoding in place.

A maintained retrieval goal gives deliberate recall a broader film-directed signal during candidate sampling. When people study mixed-category material and are asked to retrieve only one category, they can restrict recall to the target subset while temporal organization remains visible within that subset ([Lohnas et al., 2026](#); [Polyn et al., 2011](#)). In this kind of task, category-selective activity tracks retrieval effort, remains active between reported recalls, and weakens before extra-category intrusions ([Norman et al., 2017](#)). Retrieved-context models provide a formal precedent for this kind of non-temporal support: semantic, task, and source information can contribute to candidate support alongside temporal context ([Morton & Polyn, 2016](#); [Polyn et al., 2009a, 2009b](#)). Category-directed search has been formalized in associative memory models, and recent review work treats this as a natural but still underdeveloped direction for CMR ([Lohnas et al., 2026](#); [Raaijmakers & Shiffrin, 1980](#)). The simulations use this logic in reduced form by adding a maintained film-category cue during candidate sampling. This cue gives film candidates additional support without equally supporting task competitors. Task items encoded in film-overlapping context can still compete through temporal context, but they do not receive this goal-based support.

A retrieval-control process can increase film recall relative to unguided retrieval without making recall less sensitive to a reminder-linked task intervention. Protection depends on whether recall becomes less reliant on film-associated temporal context after that context has also become supportive of task material. The start-state and monitoring panels describe mechanisms that either

reinstate or preserve film-related context to drive ongoing retrieval, but since reminder-linked task encoding makes that context less selective, those mechanisms cannot substantially protect deliberate recall from the interference effect. The maintained film-retrieval goal tests that different route. It gives film candidates support from a film-specific cue, so task competitors remain available but do not receive the same added support. Under the tested conditions, that goal-based support limits the intervention’s impact enough to simulate a selective-interference effect. The simulations therefore treat increased film recall and reduced sensitivity to reminder-linked task encoding as separable outcomes.

Model Specification

Retrieved-context models implement recall as a sequence of sampling events in which current context gives graded support to candidate items (Polyn et al., 2009a; Sederberg et al., 2008). To express selective interference as changes in what gets sampled, film and non-film events have to be candidates in the same sampling process. The simulations implement this requirement by representing film, task, and delay periods as sequences of events that update context. For the film, these events correspond roughly to scenes or segments. For the task periods, they represent task items that can become competitors after reminder-linked encoding. For the delay periods, they represent filler items: non-film events that can also be encoded and later sampled. Interference manifests as a shift in sampled outcomes: task or filler candidates occupy sampling events that could otherwise retrieve film material.

The formal model links this sampling process to separate item and context representations. Each encoded event at position i is represented by an item feature vector f_i , a one-hot vector that distinguishes that event from other encoded events in the simulated paradigm. Context c_i denotes the internal context state after event i has updated context. As film, task, and filler events occur, they update context and become associated with the context active around them. During recall, the current context state determines which candidate items receive support.

During encoding, items and states of context are linked through bidirectional associations represented in two matrices. Context-to-feature memory, M^{CF} , maps context states to item

support, allowing current context to cue candidate items during recall. Feature-to-context memory, M^{FC} , maps item features to context states so that cued, reminded, or recalled items can reinstate associated context. All vectors are treated as column vectors, so $M^{FC} f_i$ is a context vector and $M^{CF} c_i$ is a feature-support vector.

Context and corresponding associations are further subdivided into temporal and source components. Temporal context distinguishes unique items so different film, task, and filler events can produce distinct context states even if items come from the same phase of the paradigm or share other features. Source context represents coarser information shared across related events, allowing recall to be shaped by task, semantic, or source relationships among items (Polyn et al., 2009a, 2009b). Emotional CMR variants distinguish emotional from neutral material in source context and give emotionally salient items stronger source-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). In the present implementation, each item also has a source feature vector s_i . Film items share a film-source feature, whereas task and filler items carry a neutral/default source feature.

Encoding the Paradigm Sequence

Film, task, and filler events become later recall candidates through context updating and item-context learning. Each event updates context and becomes associated with the resulting state. During later recall, candidate support depends on how strongly studied items are associated with the context states active during retrieval.

Before the simulated film starts, context is initialized to c_0 , a start state with no item-associated context units active. Memory matrices are initialized with pre-experimental associations that reflect baseline structure already present before the simulated paradigm begins. In M^{FC} , each item i can cue its paired context unit with weight $1 - \gamma$. In M^{CF} , each item is most strongly supported by its paired context unit, with weight δ . The other item-associated context units provide weaker background support to that item, with weight α . The start state c_0 initially

provides no item support in M^{CF} . The initial entries of M^{FC} and M^{CF} are therefore:

$$M_{pre}^{FC}(j, i) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$M_{pre}^{CF}(i, j) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases} \quad (2)$$

When event i is encountered, its representation f_i retrieves context input through feature-to-context memory:

$$c_i^{\text{IN}} = M^{FC} f_i. \quad (3)$$

Context then drifts toward the normalized input. The coefficient ρ_i keeps context normalized after the update:

$$c_i = \rho_i c_{i-1} + \beta_{\text{enc}} \hat{c}_i^{\text{IN}}. \quad (4)$$

Each event also encodes associations between its item features and the current context state. Feature-to-context learning updates M^{FC} so later cues or sampled items can reinstate associated context:

$$\Delta M^{FC} = \gamma c_i f_i^{\top}. \quad (5)$$

Context-to-feature learning updates M^{CF} so later context can support the item during sampling:

$$\Delta M^{CF} = \lambda \phi_i f_i c_i^{\top}. \quad (6)$$

Here, γ controls feature-to-context learning while λ scales context-to-feature learning. To capture primacy effects in recall, the scale ϕ_i boosts context-to-feature learning for early items in the paradigm and decays across subsequent items.

Encoding events also have a source feature vector s_i that distinguishes film items from non-film items. The source component of context follows the same normalized-drift form as temporal context, with source drift rate β_{emot} . All items can form associations with the source context active for that event, scaled by a baseline source-learning term η_s . For film encoding events, the source-context portion of the context-to-feature update receives an additional ϕ_{emot} scale to reflect stronger encoding of emotional material in the trauma-film paradigm (Cohen & Kahana, 2022; Talmi et al., 2019). Task and filler events carry the neutral/default source feature, so they use the baseline source-learning scale without the added film-source term. After encoding, film, task, and filler candidates can all enter later recall through the temporal and source-context associations learned for each item.

Reminder-Linked Task Encoding

After the film-delay period, the simulated paradigm can present film reminders before the task begins. The reminder phase reinstates film-associated context before task items are encoded. Task items encoded from that state can later receive support for recall from context that also supports film material.

The reminder phase begins by updating context toward the start state of the film sequence. Let c_b denote the context state after the break period. The reminder-phase start update is:

$$c_{r,0} = \rho_{r,0}c_b + \beta_{rem,start}c_0. \quad (7)$$

The model then applies a sequence of film-item reminder cues. Each item from the film phase is presented as a reminder cue in the order it was encoded. Feature-to-context memory retrieves reminder input as in Equation 3, and context drifts toward that input using the normalized update in Equation 4 with the reminder drift rate. Reminder cues update context but do not become recallable items and do not add new context-to-feature learning. This isolates the

reminder’s effect to reinstating film-associated context rather than adding new film-item associations.

Task events are then encoded by the ordinary event-learning rules with the reminder-updated context as the starting state. Task items carry the neutral/default source feature rather than the film-source feature, so they do not receive the added ϕ_{emot} term. To represent stronger or weaker task encoding, the simulations scale context-to-feature learning for task events by a task-encoding-strength factor m_{task} :

$$\Delta M_{\text{task}}^{CF} = m_{\text{task}} \lambda \phi_i f_i c_i^\top. \quad (8)$$

Task items become interfering competitors in later recall because they become associated with context states that can cue film material. The intervention therefore changes how non-film items are cued by film-related context, without weakening the film-item associations formed during film encoding.

Context-Guided Recall and Test-Phase Cues

In recall formats where film material must come to mind before it can be reported, the current context state assigns support to film, task, and filler candidates. At sampling step j , candidate support is computed by probing context-to-feature memory with the full context state:

$$a_{i,j} = [M^{CF} c_j]_i. \quad (9)$$

The context vector c_j includes both temporal and source-context components. Candidates already sampled in the current recall period are excluded from later sampling.

Sampling follows a Luce choice rule over the remaining candidates (Luce, 1959):

$$P(i \mid \text{continue}) = \frac{a_{i,j}^\tau}{\sum_{\text{remaining } k} a_{k,j}^\tau}. \quad (10)$$

Following likelihood-based CMR implementations, stopping is modeled as a position-dependent hazard independent of current candidate support (Kragel et al., 2015; Morton

& Polyn, 2016):

$$P_{\text{stop}}(j) = \theta_s e^{j\theta_r}. \quad (11)$$

This value is bounded to remain a valid probability. Because stopping is independent of candidate support, interference is read from the identity of sampled items. A sampled task or filler item takes the place of a possible film sample.

After a candidate is sampled, feature-to-context memory updates context toward the sampled item's associated context:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \beta_{\text{rec}} \widehat{M^{FC} f_{r_j}}. \quad (12)$$

This update lets sampled material shape later search. A sampled film item can keep later sampling near film-associated context. A sampled task or filler item can pull later sampling toward non-film material.

Test-phase film cues use the same feature-to-context operation without becoming recall outputs. When a film cue q is presented, the cued item updates context through $M^{FC} f_q$. The cue is not counted as a recalled item, does not make the cued film item unavailable for later recall, and adds no new learning. Recall sampling then resumes from the updated context state.

Deliberate Recall Control

Deliberate recall uses the same sampling rule as unguided recall, with control settings applied before, during, or after sampling. The control settings change the retrieval cue, candidate support, or sampled-item carryover without changing the film and task associations learned during encoding. This lets the simulations test which retrieval operations increase film recall and which ones protect film recall from reminder-linked task competition.

Start-of-film context reinstatement drifts context toward the state present at the beginning of film encoding before the first sampling step. The retrieval context is drifted toward the context state used to begin the film sequence, with drift rate β_{start} . This can bias early sampling toward film candidates rather than starting from the ongoing context state produced by the final delay

phase of the paradigm. This control mechanism can therefore increase film recall. However, because it still relies on film-associated context to cue candidates, it does not substantially protect deliberate recall from the interference effects of reminder-linked task encoding.

In retrieved-context models of externalized free recall, monitoring is implemented as a post-sampling match check: rejected candidates are withheld as target responses but still update the next retrieval context (Lohnas et al., 2015). The present simulations keep the post-sampling timing but change the consequence of sampled non-film candidates. Post-retrieval context monitoring reduces the context update after sampled task or filler candidates. Let $\kappa_i = 1$ for film candidates and $\kappa_i = \kappa$ for task and filler candidates. With monitoring, the sampled-item update becomes:

$$c_{j+1} = \rho_j(\kappa_i\beta_{\text{rec}})c_j + \kappa_i\beta_{\text{rec}}\widehat{M^{FC}f_{rj}}. \quad (13)$$

When $\kappa < 1$, sampled non-film candidates still count as sampled outputs but context is updated less strongly toward their associated context. This reduces carryover from sampled task or filler candidates without changing the candidate support that made them available. Like start-of-film reinstatement, this operation does not by itself remove task support from film-associated context.

The maintained film-retrieval goal is specified as a retrieval-stage film-category cue. This term abstracts the kind of semantic or category support used in retrieved-context models of selective recall, rather than updating temporal or source context during retrieval (Morton & Polyn, 2016; Polyn et al., 2011). The goal adds support to candidates marked as film items during sampling:

$$a_{i,j}^{\text{goal}} = a_{i,j} + \omega_{\text{goal}}g_i. \quad (14)$$

Here, $g_i = 1$ for film candidates and $g_i = 0$ for task and filler candidates. The indicator g_i is a retrieval-control cue, distinct from the source feature vector s_i used during encoding. Task items encoded in film-overlapping context can still compete through $a_{i,j}$, but they do not receive

the film-category increment. Therefore, unlike start-of-film reinstatement and post-sampling monitoring, the maintained film-retrieval goal can reduce the impact of reminder-linked task encoding on film recall.

The three evaluated control mechanisms therefore enter the same retrieval cycle at different points: start-state reinstatement before sampling, film-category support during sampling, and monitoring after sampling.

Supplied-Item Recognition

Recognition begins with a supplied candidate item rather than a sampled recall candidate. A presented film item retrieves associated context through the same feature-to-context associations used by reminders, cues, and sampled items. Following retrieved-context accounts of recognition, recognition evidence is the match between the context retrieved by the probe and the context active during the test (Healey & Kahana, 2016; Jin et al., 2024; Lohnas et al., 2015):

$$e_q = (M^{FC} f_q)^\top c_{\text{test}}. \quad (15)$$

The simulations use this evidence as a supplied-item access step rather than a full recognition-decision model. A recognition probe can update context after evidence is scored, but it is not treated as a recalled output. It does not enter the recallable candidate set, make the probed item unavailable for later recall, or add new learning. This implements the focal distinction between recognition and context-guided recall measures in the trauma-film paradigm: recall requires film material to be selected from context-supported candidates, whereas recognition starts with the candidate item and evaluates the context it retrieves.

Simulations

The simulations instantiate the retrieved-context account in a controlled trauma-film analogue: a laboratory task in which distressing film material is used to study memory processes relevant to responses to real-life trauma (Holmes & Bourne, 2008; James et al., 2016). The simulations are organized into three linked groups: reminder-linked task competition, retrieval-control consequences for deliberate recall, and recognition-like test-format consequences

of bidirectional item-context associations. Simulation 1 shows how a task performed after a film reminder attenuates film recall when task encoding is strong and occurs in context states that overlap with film encoding context. Simulation 2 adds the intentionality manipulation by comparing unguided retrieval with deliberate recall and then decomposing the control settings that protect film recall during deliberate retrieval. The recognition simulation compares context-to-item recall with a test format in which a candidate film item is presented for judgment.

Methodology

The simulations require default values for context drift, associative learning, retrieval choice, and stopping before the trauma-film manipulations are applied. Ideally, these values would be estimated from a selective interference experiment that implements the full paradigm modeled here: film encoding, reminder or no-reminder conditions, an interference task, a later retrieval test, and an observed dissociation between unguided and deliberate retrieval measures. Such a dataset would also need trial-by-trial recall sequences that identify which film, task, and non-film candidates entered the retrieval stream and in what order, so the model could estimate retrieval competition rather than only final film memory scores. Existing trauma-film studies define the target phenomenon, but they typically do not provide this model-fitting structure. We therefore use an immediate affective free-recall dataset as a provisional anchor for core eCMR dynamics before applying the selective interference manipulations in simulation. The anchor dataset included participants who reported concern about the war in Ukraine and studied and recalled lists of war-related and neutral photographs ([Dupertuys et al., 2026](#)). The fitted dataset included 123 participants, four lists per participant, 492 immediate recall sequences, and 3,804 scored recall events. The fit provides an externally estimated eCMR regime for deliberate context-guided recall of affective and neutral photographs. The fitted parameters set the starting encoding, retrieval, and stopping dynamics before the selective interference manipulations are applied as phase-specific encoding or retrieval-control settings. In these simulations, film items reinstate the marked source feature into source context, neutral/default events can weakly bind active source context, and retrieval begins with partial film-source reinstatement. To

accommodate the difference in total encoded items between the immediate-recall paradigm and these simulations, the primacy profile is also moderated relative to the immediate-recall fit so that the film sequence is not dominated by an unrealistically steep first-item advantage. Table 3 gives the default numerical values used in the simulations.

The simulations update associations after context integration and disallow repeated sampling of the same item. Across simulations, each condition sets phase-specific drift, learning, and retrieval-control parameters, then runs the model through encoding and retrieval. Film, break/buffer, task, and filler items are encoded by the same CMR operations defined above. When modeled as cue events, reminder cues and candidate film items presented by the test update context through Equation 3 and Equation 4 without adding new recallable items. Simulations that present candidate item information in a different test format state that separately. Each simulated trial contains 16 film items, 16 break/buffer items, 16 task items, and 16 filler items unless otherwise noted. Equal phase lengths make redistribution across film, task, and filler periods directly comparable. Retrieval allows up to 48 sampling attempts. In context-to-item recall simulations, all encoded film, break/buffer, task, and filler items can enter the retrieval competition. The simulated retrieval sequence records sampled candidates rather than only final accepted responses. This convention lets retrieval monitoring count an off-target sample as a retrieval attempt, attenuate its context update, and still allow search to continue.

A parameter sweep involves rerunning the full encode-then-retrieve simulation across planned values of a manipulated parameter. For each simulation condition and manipulated parameter value, the model first completes the phase sequence, then enters retrieval and samples recallable items using Equation 10 and Equation 11. This procedure is repeated across simulated trials within each condition/value cell. When figures plot recall across the encoded stream, they summarize retrieval after the full phase sequence has been encoded. Recall curves are estimated across simulated trials by scoring which encoded items appeared in each retrieval sequence. Serial-position curves plot the proportion of trials in which each encoding position was retrieved, and phase summaries average those proportions within phase. Thus, position and phase labels

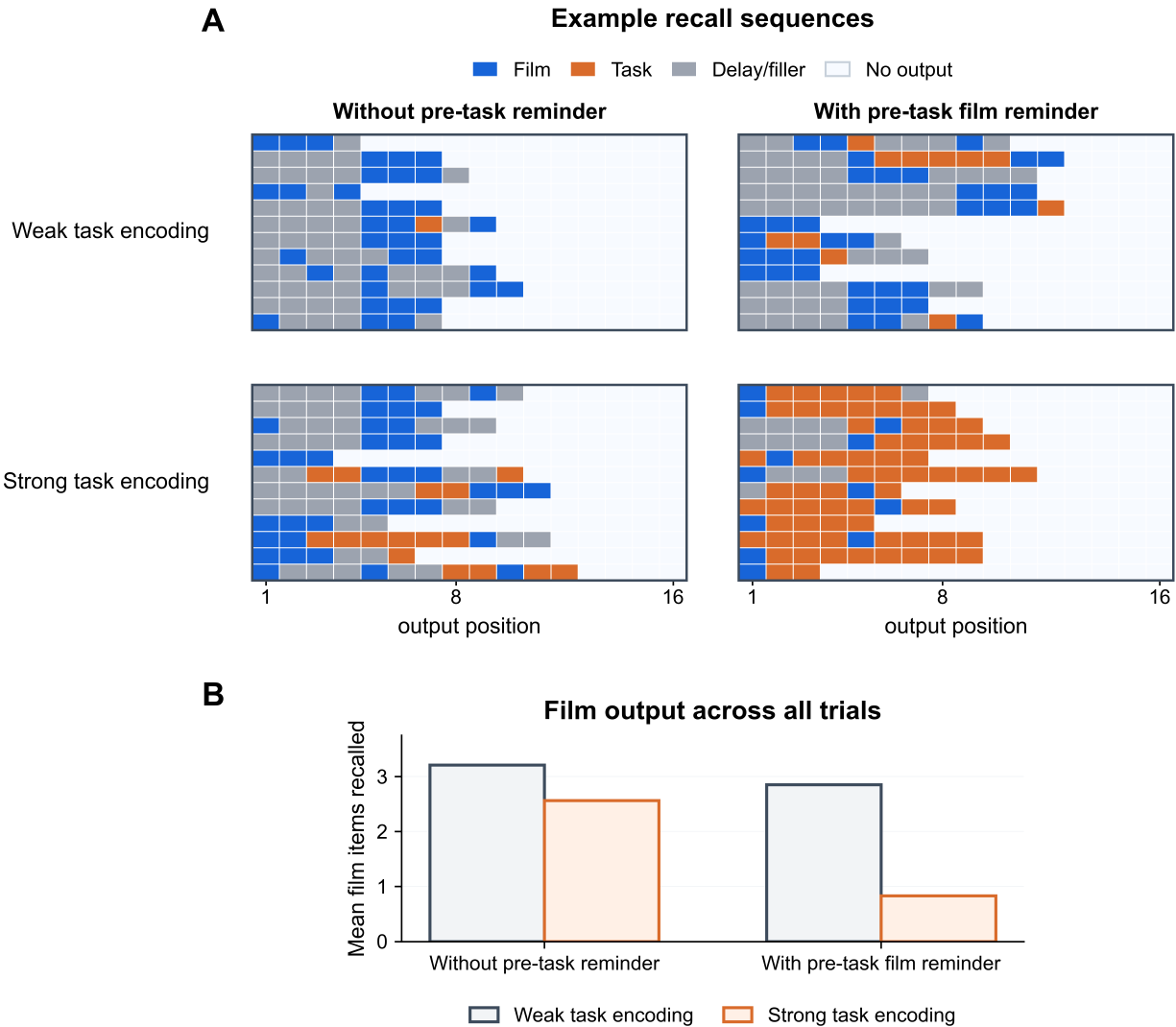
identify the encoding positions, whereas recall probabilities refer to subsequent retrieval.

Simulation 1: Task Encoding and Contextual Overlap

Simulation 1 demonstrates how a post-film task can reduce later film recall when task events are encoded in context states that overlap with the film sequence. To separate task encoding strength from reminder-driven context reinstatement, the simulation runs both the no-reminder and with-reminder conditions across the same set of task encoding strengths. We express task encoding strength as a task encoding multiplier that scales task-phase context-to-feature learning in Equation 6: 0 removes new task context-to-feature learning, 1 applies the baseline fitted rate, and values above 1 strengthen task item-context binding. In the no-reminder condition, the reminder-start and film-cue drift scales are set to zero. In the with-reminder condition, context first shifts toward the start state and then receives sequential film reminder cues before task encoding. Reminder cues update context without applying Equation 5 or Equation 6 and without adding new recallable items. Retrieval settings are held fixed, so differences in recall reflect the item-context associations formed before retrieval rather than changes in retrieval control. Because each film position is encoded with a different temporal-context state, the model predicts a serial-position pattern, not only an overall change in film recall. Task items should compete most with the film positions whose encoding contexts are most similar to the contexts active during task encoding. Simulation 1 first isolates the unguided side of the effect summarized in Figure 1. It implements the empirical task contrast as a manipulation of task encoding strength, crossed with whether a film reminder precedes task encoding. In the fixed-cue version shown in Figure 5, weak film cues are supplied periodically during recall in all conditions.

Figure 5

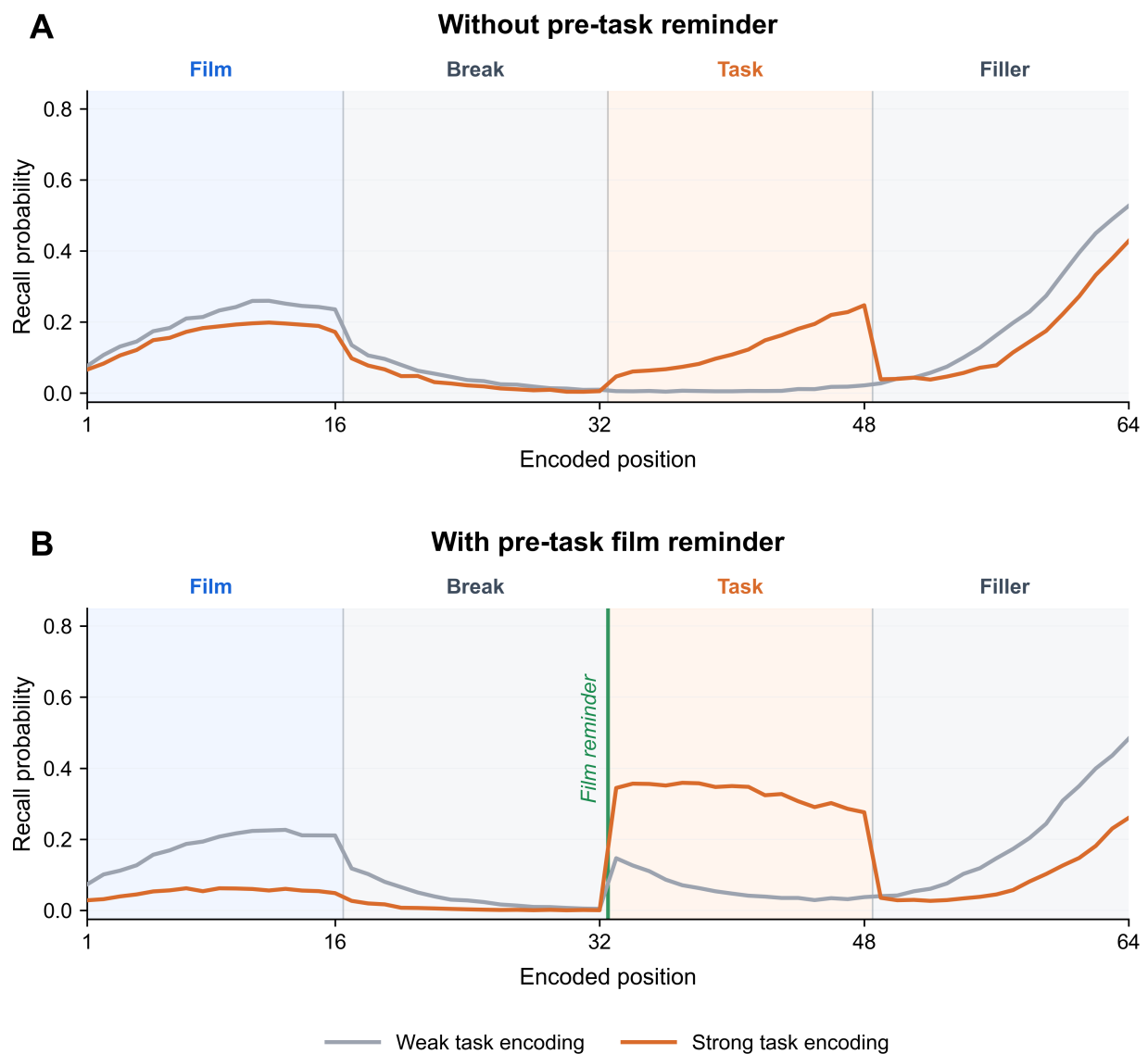
Simulation 1: unguided film recall after task encoding. Reminder condition and task encoding strength were crossed while retrieval mode was held unguided; weak film cues were supplied periodically during recall in every condition. **(A)** Example generated recall sequences for each condition. Each horizontal lane is one simulated trial selected near the condition mean for film output, with output positions running left to right. Colors mark film items, task items, delay/filler items, and unfilled output positions. **(B)** Mean number of film items recalled across all simulated trials. Film-item recall was lowest when a film reminder preceded strong task encoding, isolating the unguided-retrieval component of the selective interference effect.



These aggregate results establish the unguided interference component. The diagnostics below first show how this redistribution appears across encoded positions, then examine how it follows from context overlap; Simulation 2 then asks which deliberate-retrieval controls attenuate this loss. Table 4 summarizes the phase structure and settings used for Simulation 1. In the table, break/buffer items implement the intervening delay, and filler items implement the post-task delay. For each reminder condition and task encoding strength, serial-position curves show retrieval-test outcomes sorted by where each item appeared in the encoded sequence. Figure 7 inspects the context-overlap structure that helps explain the aggregate film and task output in Figure 5. The context-similarity analysis asks how similar each film-encoding context is to the contexts active while task items are encoded, using cosine similarity over unit-normalized temporal context states.

Figure 6

Simulation 1: reminder-linked task encoding redistributes recall probability by encoded position. Reminder condition and task encoding strength were crossed while retrieval was held unguided; weak film cues were supplied periodically during recall in all conditions. **(A)** Without a pre-task film reminder, recall probability by encoded position changes only modestly when task encoding is strengthened. **(B)** With a pre-task film reminder, strong task encoding reduces recall from film positions and increases recall from task positions. Curves compare weak and strong task encoding; shaded bands mark film, break, task, and filler positions, and the green vertical line marks the pre-task film reminder. Task encoding strength therefore changes recall probability by encoded position most clearly when task encoding follows a film reminder.



Without a pre-task film reminder, increasing the task encoding multiplier produces only modest loss of film recall and only modest gain in task recall, because task items become associated with later context states reached after drift away from the film. With a pre-task film reminder, the same task-encoding manipulation produces a larger drop in film-position recall and a larger rise in task-position recall. The condition with a pre-task film reminder therefore shows that task encoding strength matters most when task items are encoded in a film-related context state. In the selected weak-versus-strong comparison, mean film recall in the no-reminder condition changes from 3.211 to 2.562 items, while mean task recall changes from 0.143 to 2.105 items. In the condition with a pre-task film reminder, mean film recall changes from 2.852 to 0.830 items, while mean task recall changes from 0.987 to 5.282 items.

Figure 7***Simulation 1: reminder reinstatement increases overlap between film and task contexts.***

Reminder condition was varied before task encoding while retrieval was held unguided. (A) Mean task-context overlap for each encoded film position, averaging across encoded task positions. (B) Full film-by-task context-overlap matrices for the same conditions; each cell compares the context state associated with one encoded film position to the context state active at one encoded task position. Brighter cells indicate greater context overlap. The film reminder increases overlap between film-encoding and task-encoding contexts, especially for later film positions and early task positions.

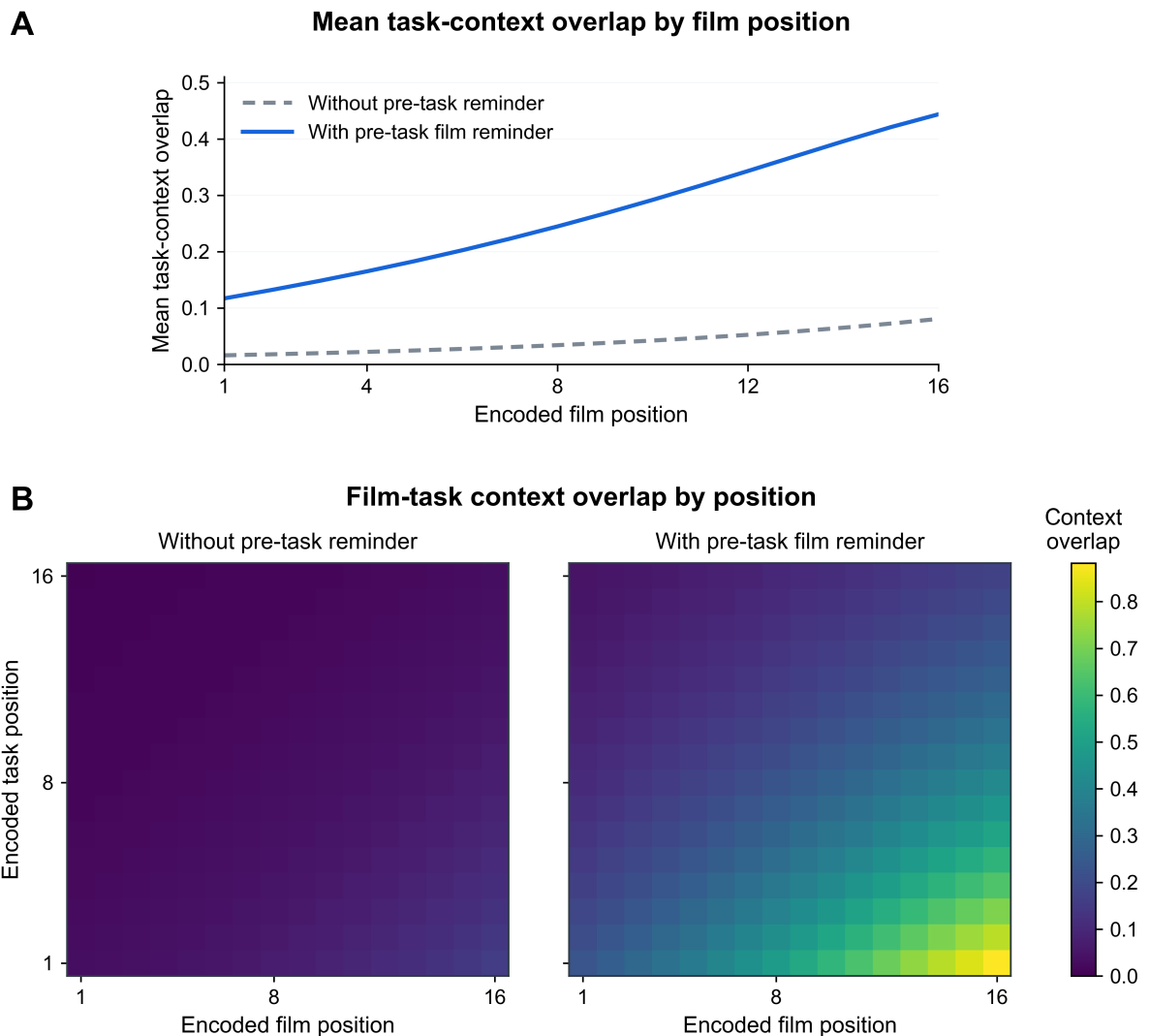


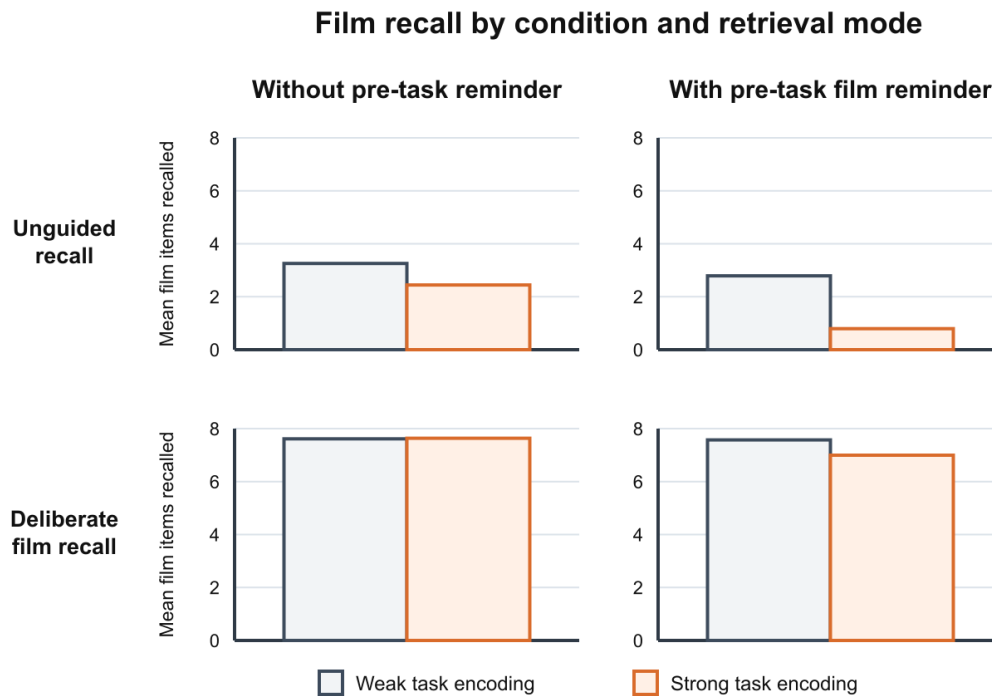
Figure 7 explains why the no-reminder and with-reminder curves diverge. Panel A summarizes temporal-context overlap between the states associated with film items and the states active during task encoding. When a sequential film reminder reinstates film-associated context before task encoding, the task-encoding contexts become more similar to some film-position contexts than to others. Panel B shows the corresponding film-position by task-position overlap matrix. Task encoding strength then determines how strongly task items are bound to these overlapping task-encoding contexts, while film encoding settings remain unchanged. The loss of film recall comes from stronger support for task items that can be cued by film-associated context. In paradigm terms, the reminder leaves the model partly oriented toward the film as task encoding begins, so task events become linked to film-like temporal context as well as the active film-source state. Task items remain distinct from film items but become increasingly likely to win retrieval from film-related cues. Simulation 2 next examines whether deliberate retrieval control reduces the impact of these task-item associations through start-of-film context reinstatement, a maintained film-retrieval goal, and retrieval monitoring.

Simulation 2: Control Mechanisms for Deliberate Recall

Simulation 2 turns from encoding conditions to retrieval control. It examines when deliberate film recall remains protected as reminder condition and task encoding strength are crossed. The deliberate-recall setting combines three control mechanisms: start-of-film context reinstatement, a maintained film-retrieval goal, and retrieval monitoring after off-target samples. Start-of-film context reinstatement drifts context toward the state present at the start of film encoding before the first retrieval attempt. The maintained film-retrieval goal adds support for film items during candidate sampling. Retrieval monitoring dampens context updating after off-target samples; in the anchor simulation, off-target context updates are partially dampened rather than fully suppressed. The figure compares this deliberate-recall setting with unguided recall across the full reminder-by-task-encoding design.

Figure 8***Simulation 2: deliberate film recall is protected from reminder-linked task interference.***

Reminder condition and task encoding strength were crossed as in Simulation 1 while retrieval mode was varied between unguided recall and deliberate film recall. Deliberate film recall combined start-of-film context reinstatement, a maintained film-retrieval goal, and moderate retrieval monitoring. Weak film cues were supplied periodically during recall in all conditions. Rows compare retrieval modes, columns compare reminder conditions, and paired bars compare weak versus strong task encoding. Strong task encoding reduced film recall most clearly when it followed a film reminder, but this reduction was much larger under unguided recall than under deliberate film recall. The model therefore simulates a selective interference effect: the reminder-plus-strong-task intervention strongly reduces unguided film access while leaving deliberate film recall relatively preserved.



In unguided recall, strengthening task encoding reduced film output modestly without a film reminder and strongly after a film reminder. In deliberate film recall, the corresponding

reduction after a film reminder was much smaller. The next figure decomposes this deliberate-retrieval setting to show which control ingredients are responsible for preserving film recall under the high-interference endpoint.

Figure 9

***Simulation 2: different retrieval controls preserve film recall to different degrees.** Each control setting was evaluated at two endpoints of the interference manipulation: one without a pre-task reminder and with weak task encoding, and one with a pre-task film reminder and with strong task encoding. Weak film cues were supplied periodically during recall in all conditions. Mean film-item recall is shown for each control setting at the two endpoints. Start-of-film reinstatement and retrieval monitoring increase film recall in some conditions, but the smallest endpoint differences occur when deliberate recall includes a film-retrieval goal.*

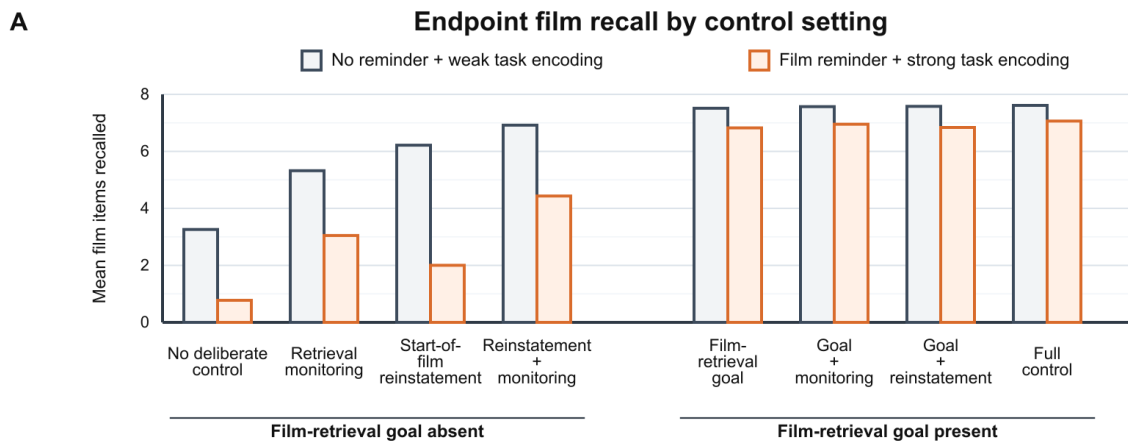


Figure 9 shows that endpoint loss remains large when the film-retrieval goal is absent. Start-of-film reinstatement and retrieval monitoring increase film recall in some conditions, yet the smallest endpoint differences occur in the goal-present conditions. The key protection comes from film-directed support that task competitors do not receive.

Figure 10

Simulation 2: a film-retrieval goal separates film-directed retrieval from task context overlap.

The high-interference condition was held fixed: a film reminder preceded strong task encoding, and weak film cues were supplied periodically during recall. A film-retrieval goal is a maintained film-category cue that supports film items during deliberate recall, distinct from the external film reminder and test-phase film cues. **(A)** Mean overlap between task-encoding contexts and film-directed retrieval cues, averaging across encoded task positions for each encoded film position. **(B)** Full overlap matrices for the same condition. The left matrix uses temporal context alone; the right matrix schematically adds the maintained film-category cue to the film-directed retrieval cue. Brighter cells indicate greater overlap with task-encoding context. Adding the maintained film-category cue reduces overlap between film-directed retrieval cues and task-encoding contexts, providing a route to film recall that depends less on temporal context remaining selective for film items.

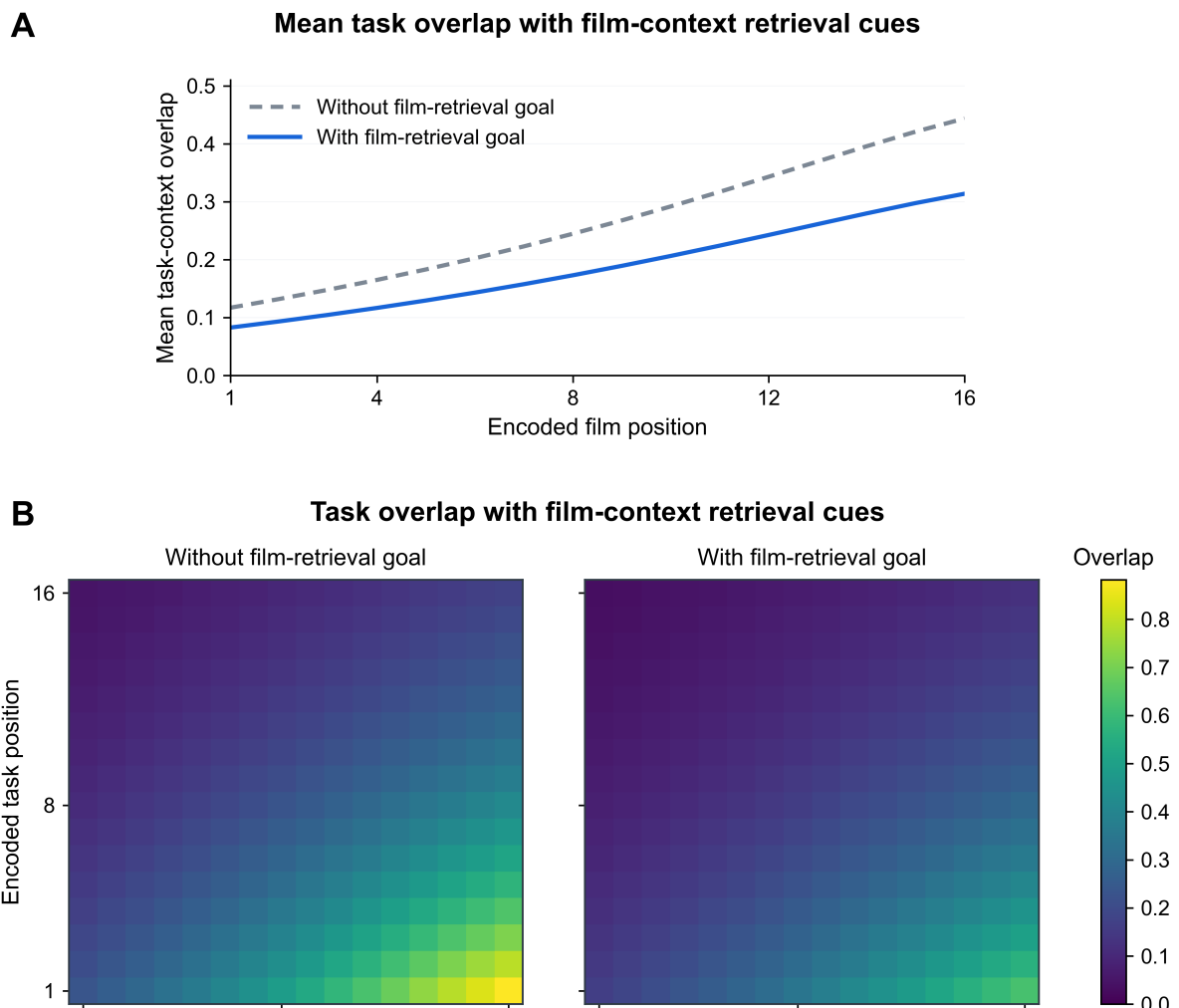
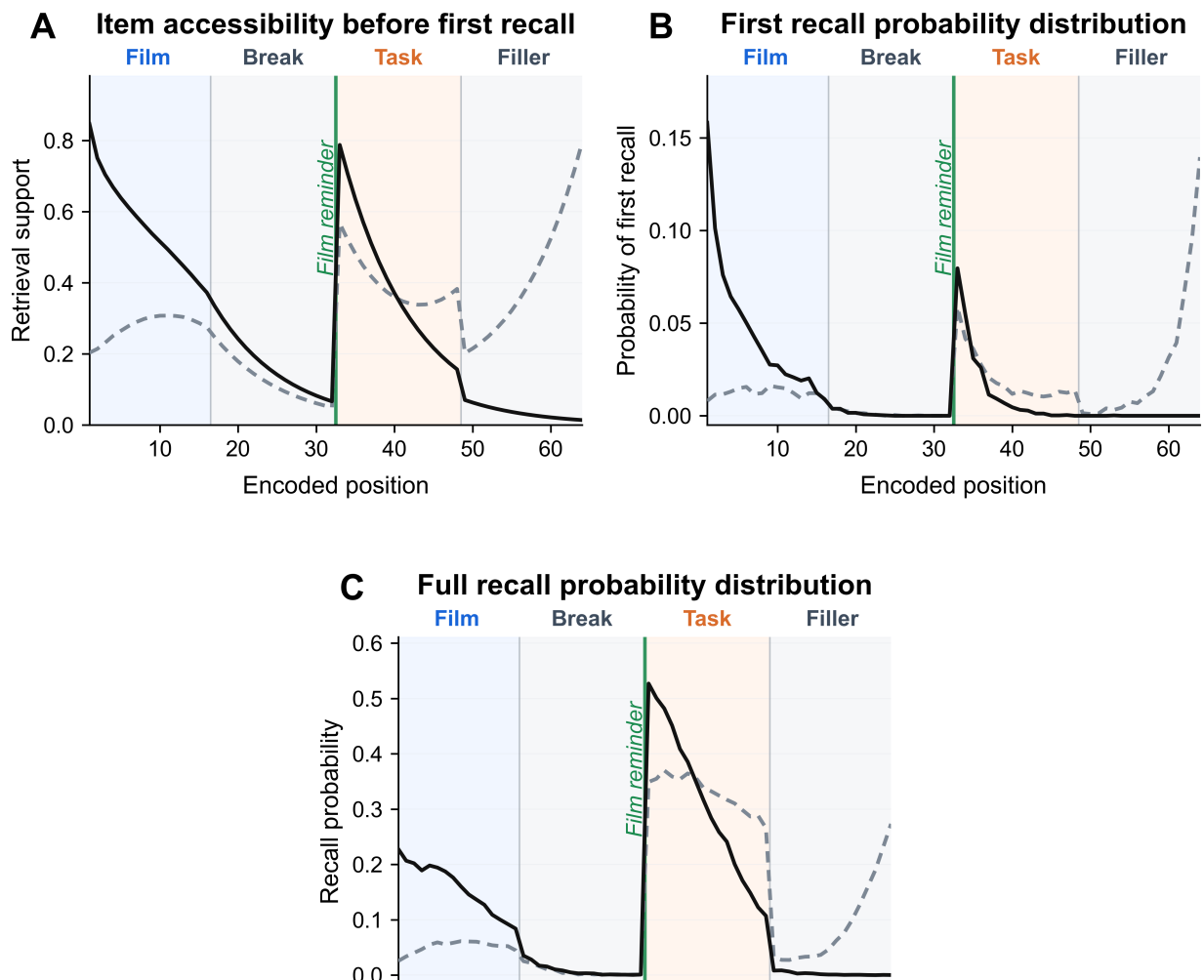


Figure Figure 10 illustrates why the goal-present conditions are less affected by reminder-linked task encoding. Adding the maintained film-category cue lowers task overlap with film-directed retrieval cues, so deliberate recall can favor film items without relying as strongly on temporal context that also supports task competitors.

The remaining diagnostics show how the other controls help without providing the same broad protection. Start-of-film context reinstatement changes where search begins, so its benefits are concentrated around early film positions.

Figure 11***Simulation 2: start-of-film context reinstatement protects early encoded film positions.***

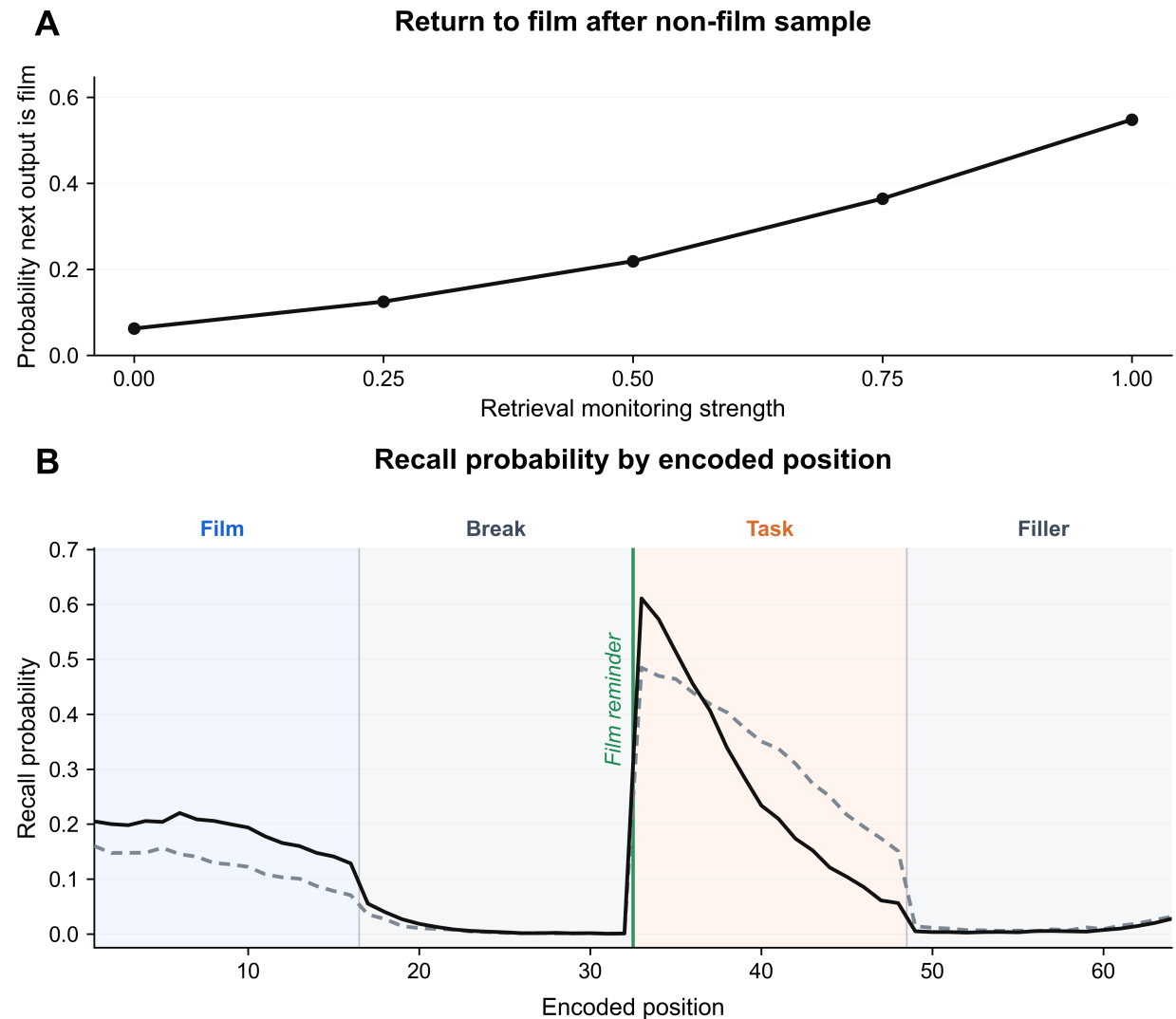
Unguided recall was compared with start-of-film context reinstatement while the condition with a pre-task film reminder and strong task encoding was held fixed and retrieval monitoring was disabled. Start-of-film context reinstatement initializes retrieval by drifting the current context toward the state present at the start of film encoding. Weak film cues were supplied periodically during recall. Shaded bands mark film, break, task, and filler positions; the green line marks the reminder immediately before task encoding. (A) Item accessibility before the first recall attempt. Reinstatement increases accessibility most strongly for early encoded film positions. (B) First recall probability distribution. Reinstatement shifts recall initiation toward early film positions. (C) Full recall probability distribution. The early-film bias remains visible across the recall period, while later film positions remain more vulnerable under reminder-linked strong task encoding. Start-of-film context reinstatement therefore protects early encoded film positions by shifting retrieval initiation toward the start of the film sequence.



Retrieval monitoring addresses a later part of the retrieval sequence. When search samples a non-film item, monitoring limits how strongly that sample redirects subsequent retrieval.

Figure 12

Simulation 2: retrieval monitoring supports return to film recall. The condition with a pre-task film reminder and strong task encoding was held fixed, start-of-film context reinstatement was present, and retrieval monitoring strength was varied. Retrieval monitoring dampens context updating after non-film samples during deliberate recall. Weak film cues were supplied periodically during recall. **(A)** Model-diagnostic probability that a non-film sample was followed by a film output as retrieval monitoring increased. **(B)** Recall probability by encoded position with no retrieval monitoring versus the moderate retrieval-monitoring setting used in the Simulation 2 anchor figure. Shaded bands mark film, break, task, and filler positions; the green line marks the reminder immediately before task encoding. Retrieval monitoring therefore supports film recall after search is pulled away from the film sequence, increasing film recall probability under reminder-linked strong task encoding.



Recognition-Like Test Format

Recognition tests in the trauma-film literature present candidate film stills or statements and ask participants to judge whether they appeared in the film ([Lau-Zhu et al., 2021](#)). Retrieved-context models have also been extended to recognition more generally: a test probe can reinstate its associated context, and the match between reinstated context and the current retrieval context can contribute to an old-new decision ([Healey & Kahana, 2016](#); [Kahana, 2020](#)). Related recognition findings show that temporal context retrieved by one probe can influence recognition of nearby studied probes ([Schwartz et al., 2005](#)). Here, the model isolates the access step that recognition provides. The candidate film item is available first and can reinstate associated context before any need for that item to win a context-to-item recall competition. This analysis does not model decision criteria, confidence, or complete foil processing. It asks whether presenting the candidate item changes where reminder-linked task competition enters retrieval.

Figure 13

Item-to-context retrieval bypasses reminder-linked task interference. (A) CMR stores bidirectional associations between item features and context. Feature-to-context memory lets an available item reinstate its associated context, whereas context-to-feature memory lets context cue candidate items. (B) When film-associated context cues items, film items and reminder-linked task items can both receive support. When a film item reinstates context, retrieval targets context units instead, so task items are not competitors in that step. (C) In recognition, the tested film item is already available. The item retrieves associated context, which is compared with ongoing test context; stronger matches provide stronger evidence that the item appeared in the film. (D) Film recognition accuracy is plotted across the same reminder and task-encoding conditions used in the recall simulations. Accuracy is hit rate minus false-alarm rate for unstudied film-like lures. The reminder-plus-strong-task condition does not produce a corresponding reduction in recognition accuracy.

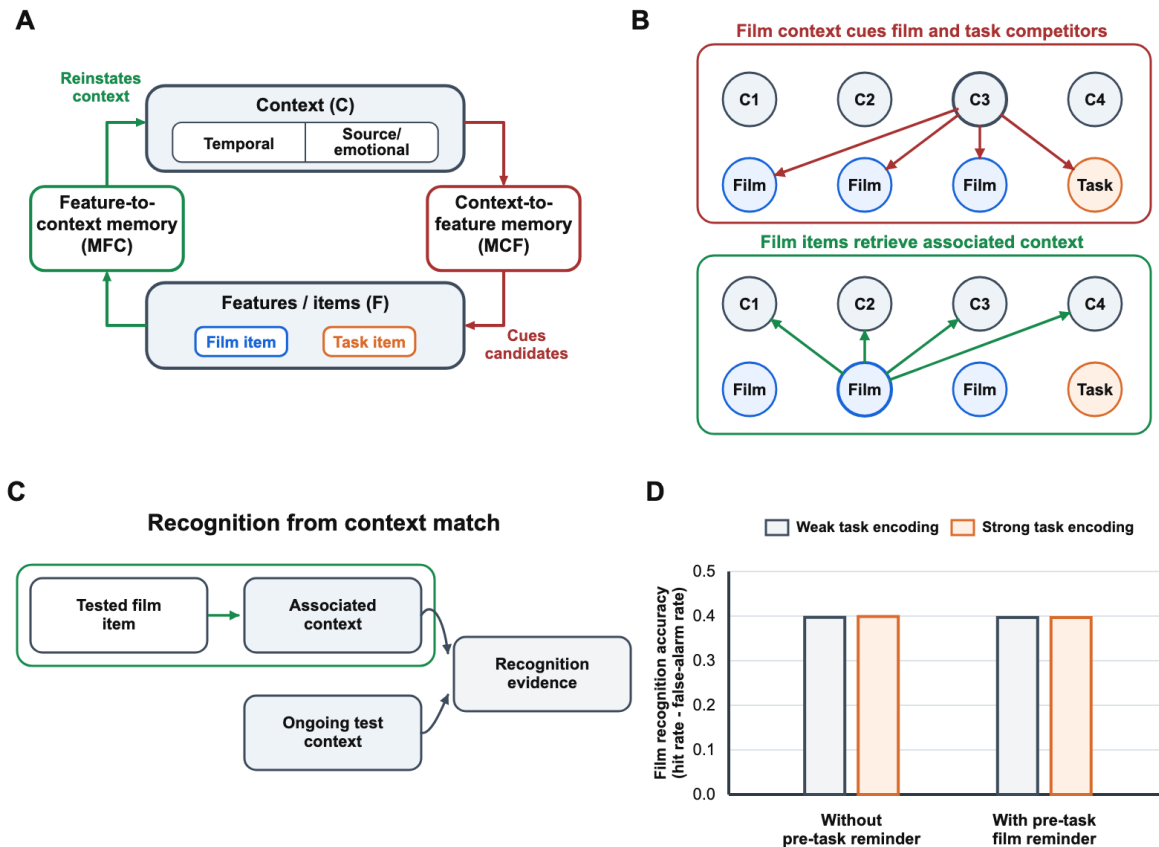


Figure 13 shows the resulting test-format consequence. In context-guided recall, reminder-linked task items can compete when the current context supports both film and task candidates. In recognition-like tests, the presented candidate item reinstates context directly. The task items created by reminder-linked encoding therefore have less opportunity to compete at the initial access step, even though the same learned associations are present in memory.

General Discussion

References

- Asselbergs, J., Riper, H., Engelhard, I. M., Mannes, F., & Sijbrandij, M. (2024). The effectiveness of two novel approaches to prevent intrusions: A pilot study comparing Tetris_dualtask and imagery rescripting to control. *Journal of Behavior Therapy and Experimental Psychiatry*, 82, 101920. <https://doi.org/10.1016/j.jbtep.2023.101920>
- Asselbergs, J., van Bentum, J., Riper, H., Cuijpers, P., Holmes, E. A., & Sijbrandij, M. (2023). A systematic review and meta-analysis of the effect of cognitive interventions to prevent intrusive memories using the trauma film paradigm. *Journal of Psychiatric Research*, 159, 116–129. <https://doi.org/10.1016/j.jpsychires.2023.01.028>
- Baddeley, A. D., & Andrade, J. (2000). Working memory and the vividness of imagery. *Journal of Experimental Psychology: General*, 129(1), 126–145. <https://doi.org/10.1037/0096-3445.129.1.126>
- Beckenstrom, A. C., Bonsall, M. B., Markham, A., Slade, O., Ramineni, V., Iyadurai, L., Islam, Z., Highfield, J., Jaki, T., Goodwin, G. M., Dias, R., Daniels, R., Malik, A., Summers, C., Kingslake, J., & Holmes, E. A. (2026). A digital imagery-competing task intervention for stopping intrusive memories in trauma-exposed health-care staff during the COVID-19 pandemic in the UK: A bayesian adaptive randomised clinical trial. *The Lancet Psychiatry*, 13(3), 233–247. [https://doi.org/10.1016/s2215-0366\(25\)00397-9](https://doi.org/10.1016/s2215-0366(25)00397-9)
- Brewin, C. R. (2014). Episodic memory, perceptual memory, and their interaction: Foundations for a theory of posttraumatic stress disorder. *Psychological Bulletin*, 140(1), 69–97. <https://doi.org/10.1037/a0033722>

Brewin, C. R., Dalgleish, T., & Joseph, S. (1996). A dual representation theory of posttraumatic stress disorder. *Psychological Review*, 103(4), 670–686.

<https://doi.org/10.1037/0033-295X.103.4.670>

Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications. *Psychological Review*, 117(1), 210–232. <https://doi.org/10.1037/a0018113>

Busemeyer, J. R., & Wang, Y.-M. (2000). Model comparisons and model selections based on generalization criterion methodology. *Journal of Mathematical Psychology*, 44(1), 171–189. <https://doi.org/10.1006/jmps.1999.1282>

Cohen, R. T., & Kahana, M. J. (2022). A memory-based theory of emotional disorders. *Psychological Review*, 129(4), 742–776. <https://doi.org/10.1037/rev0000334>

Deepröse, C., Zhang, S., DeJong, H., Dalgleish, T., & Holmes, E. A. (2012). Imagery in the aftermath of viewing a traumatic film: Using cognitive tasks to modulate the development of involuntary memory. *Journal of Behavior Therapy and Experimental Psychiatry*, 43(2), 758–764. <https://doi.org/10.1016/j.jbtep.2011.10.008>

Dupertuys, J., Shulhan, A., & Talmi, D. (2026). *Ukraine Memory experiment data set*.

Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319–345. [https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/10.1016/S0005-7967(99)00123-0)

Healey, M. K., & Kahana, M. J. (2014). Is memory search governed by universal principles or idiosyncratic strategies? *Journal of Experimental Psychology: General*, 143(2), 575–596. <https://doi.org/10.1037/a0033715>

Healey, M. K., & Kahana, M. J. (2016). A four-component model of age-related memory change. *Psychological Review*, 123(1), 23–69. <https://doi.org/10.1037/rev0000015>

Healey, M. K., & Wahlheim, C. N. (2024). PEPPR: A post-encoding pre-production reinstatement model of dual-list free recall. *Memory & Cognition*, 52(1), 163–181. <https://doi.org/10.3758/s13421-023-01453-z>

Hemi, A., Sopp, M. R., Perel, A., Holmes, E. A., & Levy-Gigi, E. (2023). Cognitive flexibility

moderates the efficacy of a visuospatial intervention following exposure to analog trauma.

Journal of Behavior Therapy and Experimental Psychiatry, 81, 101858.

<https://doi.org/10.1016/j.jbtep.2023.101858>

Herrema, H. G., & Kahana, M. J. (2024). *First recall costs and benefits*. PsyArXiv.

<https://doi.org/10.31234/osf.io/sm3fw>

Holmes, E. A., & Bourne, C. (2008). Inducing and modulating intrusive emotional memories: A review of the trauma film paradigm. *Acta Psychologica*, 127(3), 553–566.

<https://doi.org/10.1016/j.actpsy.2007.11.002>

Holmes, E. A., Brewin, C. R., & Hennessey, R. G. (2004). Trauma films, information processing, and intrusive memory development. *Journal of Experimental Psychology: General*, 133(1), 3–22. <https://doi.org/10.1037/0096-3445.133.1.3>

Holmes, E. A., James, E. L., Coode-Bate, T., & Deeproose, C. (2009). Can playing the computer game Tetris reduce the build-up of flashbacks for trauma? A proposal from cognitive science. *PLoS ONE*, 4(1), e4153. <https://doi.org/10.1371/journal.pone.0004153>

Holmes, E. A., James, E. L., Kilford, E. J., & Deeproose, C. (2010). Key steps in developing a cognitive vaccine against traumatic flashbacks: Visuospatial Tetris versus verbal Pub Quiz. *PLoS ONE*, 5(11), e13706. <https://doi.org/10.1371/journal.pone.0013706>

Horsch, A., Vial, Y., Favrod, C., Harari, M. M., Blackwell, S. E., Watson, P., Iyadurai, L., Bonsall, M. B., & Holmes, E. A. (2017). Reducing intrusive traumatic memories after emergency caesarean section: A proof-of-principle randomized controlled study. *Behaviour Research and Therapy*, 94, 36–47. <https://doi.org/10.1016/j.brat.2017.03.018>

Howard, M. W., & Kahana, M. J. (1999). Contextual variability and serial position effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25(4), 923–941. <https://doi.org/10.1037/0278-7393.25.4.923>

Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299.

<https://doi.org/10.1006/jmps.2001.1388>

Howard, M. W., Youker, T. E., & Venkatadass, V. S. (2008). The persistence of memory:

Contiguity effects across hundreds of seconds. *Psychonomic Bulletin & Review*, 15(1), 58–63.

Hupbach, A., Hardt, O., Gomez, R., & Nadel, L. (2008). The dynamics of memory:

Context-dependent updating. *Learning & Memory*, 15(8), 574–579.

<https://doi.org/10.1101/lm.1022308>

Iyadurai, L., Blackwell, S. E., Meiser-Stedman, R., Watson, P. C., Bonsall, M. B., Geddes, J. R.,

Nobre, A. C., & Holmes, E. A. (2018). Preventing intrusive memories after trauma via a brief intervention involving Tetris computer game play in the emergency department: A proof-of-concept randomized controlled trial. *Molecular Psychiatry*, 23(3), 674–682.

<https://doi.org/10.1038/mp.2017.23>

Iyadurai, L., Highfield, J., Kanstrup, M., Markham, A., Ramineni, V., Guo, B., Jaki, T.,

Kingslake, J., Goodwin, G. M., Summers, C., Bonsall, M. B., & Holmes, E. A. (2023).

Reducing intrusive memories after trauma via an imagery-competing task intervention in COVID-19 intensive care staff: A randomised controlled trial. *Translational Psychiatry*, 13(1), 290. <https://doi.org/10.1038/s41398-023-02578-0>

Iyadurai, L., Visser, R. M., Lau-Zhu, A., Porcheret, K., Horsch, A., Holmes, E. A., & James, E. L.

(2019). Intrusive memories of trauma: A target for research bridging cognitive science and its clinical application. *Clinical Psychology Review*, 69, 67–82.

<https://doi.org/10.1016/j.cpr.2018.08.005>

James, E. L., Bonsall, M. B., Hoppitt, L., Tunbridge, E. M., Geddes, J. R., Milton, A. L., &

Holmes, E. A. (2015). Computer game play reduces intrusive memories of experimental trauma via reconsolidation-update mechanisms. *Psychological Science*, 26(8), 1201–1215.

<https://doi.org/10.1177/0956797615583071>

James, E. L., Lau-Zhu, A., Clark, I. A., Visser, R. M., Hageraars, M. A., & Holmes, E. A. (2016).

The trauma film paradigm as an experimental psychopathology model of psychological trauma: Intrusive memories and beyond. *Clinical Psychology Review*, 47, 106–142.

<https://doi.org/10.1016/j.cpr.2016.04.010>

- Jin, B. J., Halpern, D. J., & Kahana, M. J. (2024). *A theory of memory for items and associations*.
<https://memory.psych.upenn.edu/files/pubs/JinEtal24.pdf>
- Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1), 103–109.
- Kahana, M. J. (2020). Computational models of memory search. *Annual Review of Psychology*, 71, 107–138. <https://doi.org/10.1146/annurev-psych-010418-103358>
- Kanstrup, M., Kontio, E., Geranmayeh, A., Olofsdotter Lauri, K., Moulds, M. L., & Holmes, E. A. (2021). A single case series using visuospatial task interference to reduce the number of visual intrusive memories of trauma with refugees. *Clinical Psychology & Psychotherapy*, 28(1), 109–123. <https://doi.org/10.1002/cpp.2489>
- Kanstrup, M., Singh, L., Leehr, E. J., Göransson, K. E., Pihlgren, S. A., Iyadurai, L., Dahl, O., Falk, A.-C., Lindström, V., Hadziosmanovic, N., Gabrysch, K., Moulds, M. L., & Holmes, E. A. (2024). A guided single session intervention to reduce intrusive memories of work-related trauma: A randomised controlled trial with healthcare workers in the COVID-19 pandemic. *BMC Medicine*, 22(1), 403. <https://doi.org/10.1186/s12916-024-03569-8>
- Kessler, H., Schmidt, A.-C., James, E. L., Blackwell, S. E., von Rauchhaupt, M., Harren, K., Kehyayan, A., Clark, I. A., Sauvage, M., Herpertz, S., Axmacher, N., & Holmes, E. A. (2020). Visuospatial computer game play after memory reminder delivered three days after a traumatic film reduces the number of intrusive memories of the experimental trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 67, 101454.
<https://doi.org/10.1016/j.jbtep.2019.01.006>
- Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. *Nature Neuroscience*, 12(3), 256–258.
<https://doi.org/10.1038/nn.2271>
- Kragel, J. E., Morton, N. W., & Polyn, S. M. (2015). Neural activity in the medial temporal lobe reveals the fidelity of mental time travel. *Journal of Neuroscience*, 35(7), 2914–2926.
<https://doi.org/10.1523/JNEUROSCI.3378-14.2015>

- Krans, J., Näring, G., Becker, E. S., & Holmes, E. A. (2009). Intrusive trauma memory: A review and functional analysis. *Applied Cognitive Psychology*, 23(8), 1076–1088.
<https://doi.org/10.1002/acp.1611>
- Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2019). Intrusive memories and voluntary memory of a trauma film: Differential effects of a cognitive interference task after encoding. *Journal of Experimental Psychology: General*, 148(12), 2154–2180. <https://doi.org/10.1037/xge0000598>
- Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2021). Selectively interfering with intrusive but not voluntary memories of a trauma film: Accounting for the role of associative memory. *Clinical Psychological Science*, 9(6), 1128–1143. <https://doi.org/10.1177/2167702621998315>
- Lohnas, L. J., Healey, M. K., & Davachi, L. (2023). Neural temporal context reinstatement of event structure during memory recall. *Journal of Experimental Psychology: General*, 152(7), 1840–1872. <https://doi.org/10.1037/xge0001354>
- Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363.
<https://doi.org/10.1037/a0039036>
- Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2026). Specialized recall procedures. *Psychonomic Bulletin & Review*, 33(3), 96. <https://doi.org/10.3758/s13423-025-02744-z>
- Luce, R. D. (1959). *Individual choice behavior: A theoretical analysis*. Wiley.
- McConnell, T. G. W., Bernhard, H., Holmes, E. A., Henson, R., & Talmi, D. (2026). *Testing the dual representation account of memory for trauma*. PsyArXiv preprint.
https://osf.io/preprints/psyarxiv/m3pnz_v1/
- Morton, N. W., & Polyn, S. M. (2016). A predictive framework for evaluating models of semantic organization in free recall. *Journal of Memory and Language*, 86, 119–140.
<https://doi.org/10.1016/j.jml.2015.10.002>
- Norman, Y., Yeagle, E. M., Harel, M., Mehta, A. D., & Malach, R. (2017). Neuronal baseline shifts underlying boundary setting during free recall. *Nature Communications*, 8(1), 1301.
<https://doi.org/10.1038/s41467-017-01184-1>

- Polyn, S. M., Erlichman, G., & Kahana, M. J. (2011). Semantic cuing and the scale insensitivity of recency and contiguity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37(3), 766–775. <https://doi.org/10.1037/a0022475>
- Polyn, S. M., Kragel, J. E., Morton, N. W., McCluey, J. D., & Cohen, Z. D. (2012). The neural dynamics of task context in free recall. *Neuropsychologia*, 50(4), 447–457. <https://doi.org/10.1016/j.neuropsychologia.2011.08.025>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009a). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156. <https://doi.org/10.1037/a0014420>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009b). Task context and organization in free recall. *Neuropsychologia*, 47(11), 2158–2163. <https://doi.org/10.1016/j.neuropsychologia.2009.02.013>
- Raaijmakers, J. G. W., & Shiffrin, R. M. (1980). SAM: A theory of probabilistic search of associative memory. In G. H. Bower (Ed.), *The psychology of learning and motivation: Advances in research and theory* (Vol. 14, pp. 207–262). Academic Press. [https://doi.org/10.1016/S0079-7421\(08\)60162-0](https://doi.org/10.1016/S0079-7421(08)60162-0)
- Ramineni, V., Millroth, P., Iyadurai, L., Jaki, T., Kingslake, J., Highfield, J., Summers, C., Bonsall, M. B., & Holmes, E. A. (2023). Treating intrusive memories after trauma in healthcare workers: A bayesian adaptive randomised trial developing an imagery-competing task intervention. *Molecular Psychiatry*, 28(7), 2985–2994. <https://doi.org/10.1038/s41380-023-02062-7>
- Schwartz, G., Howard, M. W., Jing, B., & Kahana, M. J. (2005). Shadows of the past: Temporal retrieval effects in recognition memory. *Psychological Science*, 16(11), 898–904. <https://doi.org/10.1111/j.1467-9280.2005.01634.x>
- Sederberg, P. B., Gershman, S. J., Polyn, S. M., & Norman, K. A. (2011). Human memory reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin & Review*, 18(3), 455–468. <https://doi.org/10.3758/s13423-011-0086-9>

- Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912.
<https://doi.org/10.1037/a0013396>
- Talmi, D. (2013). Enhanced emotional memory: Cognitive and neural mechanisms. *Current Directions in Psychological Science*, 22(6), 430–436.
<https://doi.org/10.1177/0963721413498893>
- Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485.
<https://doi.org/10.1037/rev0000132>
- Talmi, D., Luk, B. T. C., McGarry, L. M., & Moscovitch, M. (2007). The contribution of relatedness and distinctiveness to emotionally-enhanced memory. *Journal of Memory and Language*, 56(4), 555–574. <https://doi.org/10.1016/j.jml.2007.01.002>
- Talmi, D., & Moscovitch, M. (2004). Can semantic relatedness explain the enhancement of memory for emotional words? *Memory & Cognition*, 32(5), 742–751.
<https://doi.org/10.3758/bf03195864>
- Unsworth, N., Brewer, G. A., & Spillers, G. J. (2010). Understanding the dynamics of correct and error responses in free recall: Evidence from externalized free recall. *Memory & Cognition*, 38(4), 419–430. <https://doi.org/10.3758/MC.38.4.419>
- Varma, M. M., Zeng, S., Singh, L., Holmes, E. A., Huang, J., Chiu, M. H., & Hu, X. (2024). A systematic review and meta-analysis of experimental methods for modulating intrusive memories following lab-analogue trauma exposure in non-clinical populations. *Nature Human Behaviour*, 8(10), 1968–1987. <https://doi.org/10.1038/s41562-024-01956-y>
- Wessel, I., Krans, J., Albers, C. J., Chauhan, N., Ferreira de Sá, D. S., Hauck, A., Jaswetz, L., Michael, T., Nixon, R. D. V., Pearson, D. G., Slattery, I., Smeets, T., Takarangi, M. K. T., Willems, T. R. A., & van Schie, K. (2025). Evidence that Tetris reduces immediate but not subsequent daily intrusions of a trauma film: A multilab replication study. *Collabra: Psychology*, 11(1), 130791. <https://doi.org/10.1525/collabra.130791>

Zaromb, F. M., Howard, M. W., Dolan, E. D., Sirotin, Y. B., Tully, M., Wingfield, A., & Kahana, M. J. (2006). Temporal associations and prior-list intrusions in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(4), 792–804.

<https://doi.org/10.1037/0278-7393.32.4.792>

Zhou, C. Y., Talmi, D., Daw, N. D., & Mattar, M. G. (2025). Episodic retrieval for model-based evaluation in sequential decision tasks. *Psychological Review*, 132(1), 18–49.

<https://doi.org/10.1037/rev0000505>

Table 1

Core notation used in the model specification. Symbols used only in one operation are defined when that operation is specified, and numerical values are reported with the simulation methodology.

Symbol	Role	Main use
f_i	Item feature vector for encoded event i	Encoding, reminders, recall updates, recognition probes
s_i	Source feature vector distinguishing film from non-film items	Source-context encoding
c_i	Context state after encoded event i	Encoding and learning
M^{FC}	Feature-to-context associations	Context reinstatement from encoded, cued, sampled, or probed items
M^{CF}	Context-to-feature associations	Candidate support during context-guided recall
α	Background support from nonpaired item-associated context units	Initial M^{CF}
δ	Support from each item's paired context unit	Initial M^{CF}
β_{enc}	Drift toward item-cued context input during encoding	Film, task, and filler encoding
β_{rec}	Drift toward a sampled item's associated context	Recall updates
β_{emot}	Drift in the source-context component	Source-context encoding
γ	Feature-to-context learning rate	Updates to M^{FC}

Table 2*Overview of what is held fixed, varied, and measured across simulations.*

Simulation	Main fixed structure	Varied or compared	Primary readout
1	Unguided retrieval settings; film, delay, task, and filler phases	No-reminder versus with-reminder task encoding; task encoding multiplier	Film and task recall by encoded position; film-task context overlap
2	Reminder condition and task encoding strength crossed as in Simulation 1	Unguided retrieval versus deliberate recall; start-of-film context reinstatement, maintained film-retrieval goal, and retrieval monitoring	Film recall across retrieval modes; endpoint differences across control settings; recall initiation and film return after off-target samples
Recognition	Same reminder and task-encoding conditions used in recall simulations, without a full recognition-decision model	Context-to-item recall versus candidate film items presented for judgment	Whether item-to-context access changes where interference enters retrieval

Table 3

Default numerical parameter values. Values are from a pooled eCMR fit to the immediate free-recall sequences from Dupertuys et al. (2026), with α fixed to zero, except for the source-context and primacy settings described in the text.

Symbol	Value	Role
β_{enc}	0.440	Encoding drift rate
β_{start}	0.766	Start drift rate
β_{rec}	0.678	Recall drift rate
γ	0.468	Feature-to-context learning rate
δ	0.00022	Pre-experimental self-support
α	0	Pre-experimental shared support
ϕ_s	0.100	Primacy scale used in simulations
ϕ_d	1.500	Primacy decay used in simulations
τ	3.562	Choice sensitivity
β_{emot}	0.696	Source-context drift rate
ϕ_{emot}	0.130	Learning scale for the marked source feature
η_s	0.050	Baseline source-context-to-feature learning
$\beta_{film-src}$	0.500	Fixed film-source reinstatement at retrieval onset
θ_s	0.0123	Stop scale
θ_r	0.372	Stop growth

Table 4

Phase structure and settings for Simulation 1. Scale values multiply the corresponding base rates or learning strengths.

Phase or setting	Current value	Role
Film items	16	Target episode
Break/buffer items	16	Intervening non-film experience before reminder/task
Film reminder cues	16	Sequential film cues, no new learning
Task items	16	Potential competitors
Filler items	16	Post-task non-film experience
Maximum sampling attempts	48	Upper bound on sampled retrieval attempts
Simulated trials per condition	5000	For each reminder condition and task encoding multiplier
Break and filler drift/learning scales	1.0	Baseline non-film encoding settings
No-reminder drift scales	0.0	Omits reminder-start and reminder-cue reinstatement
With-reminder drift scales	1.0	Applies reminder-start and film cue reinstatement
Task encoding multiplier	1.0 or 2.0	Multiplies task-phase context-to-feature learning during task encoding
Film source marking	On	Film items carry the marked source feature
Task source marking	Off	Task items use the